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(54) **BUILDING PERFORMANCE ASSESSMENT
SYSTEM AND METHOD**

(71) Applicant: **The AUROS Group, Inc.**, Carnegie,
PA (US)

(72) Inventors: **Beth Ann Eckenrode**, Presto, PA (US);
Craig Enrico Stevenson, Oakdale, PA
(US)

(73) Assignee: **The AUROS Group, Inc.**, Carnegie,
PA (US)

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G01M 99/00 (2011.01)
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(52) **U.S. Cl.**
CPC **G06F 30/13** (2020.01); **G01M 99/00**
(2013.01); **G06Q 10/06315** (2013.01); **G06Q**
10/067 (2013.01); **G06F 2111/08** (2020.01)

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G06Q 10/06315

See application file for complete search history.

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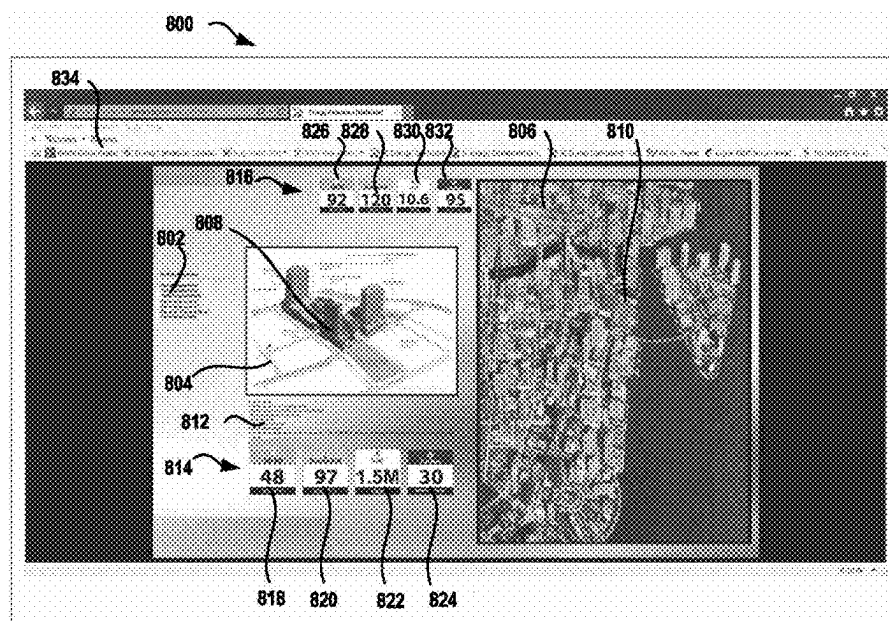
Primary Examiner — Bijan Mapar

(74) *Attorney, Agent, or Firm* — Bauer and Joseph;
Thomas Joseph

(57) **ABSTRACT**

A virtual data acquisition component holds a physics-based,
building energy model having a plurality of predicted building
performance metrics produced through a simulation of
expected building performance. A physical data acquisition
component obtains a plurality of trended building performance
metrics during the operation of a building. An integrated
interface having an analytics platform receives the plurality of
predicted building performance metrics from the
virtual data acquisition component and the plurality of
trended building performance metrics from the physical data
acquisition component and produces an analytic data product.
The analytic data product can be compared to the
predicted building performance metrics or the trended building
performance metrics.

20 Claims, 21 Drawing Sheets



Related U.S. Application Data

which is a continuation of application No. 15/907,222, filed on Feb. 27, 2018, now Pat. No. 10,956,627.

- (60) Provisional application No. 62/530,563, filed on Jul. 10, 2017.

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G06Q 10/0631 (2023.01)
G06Q 10/067 (2023.01)

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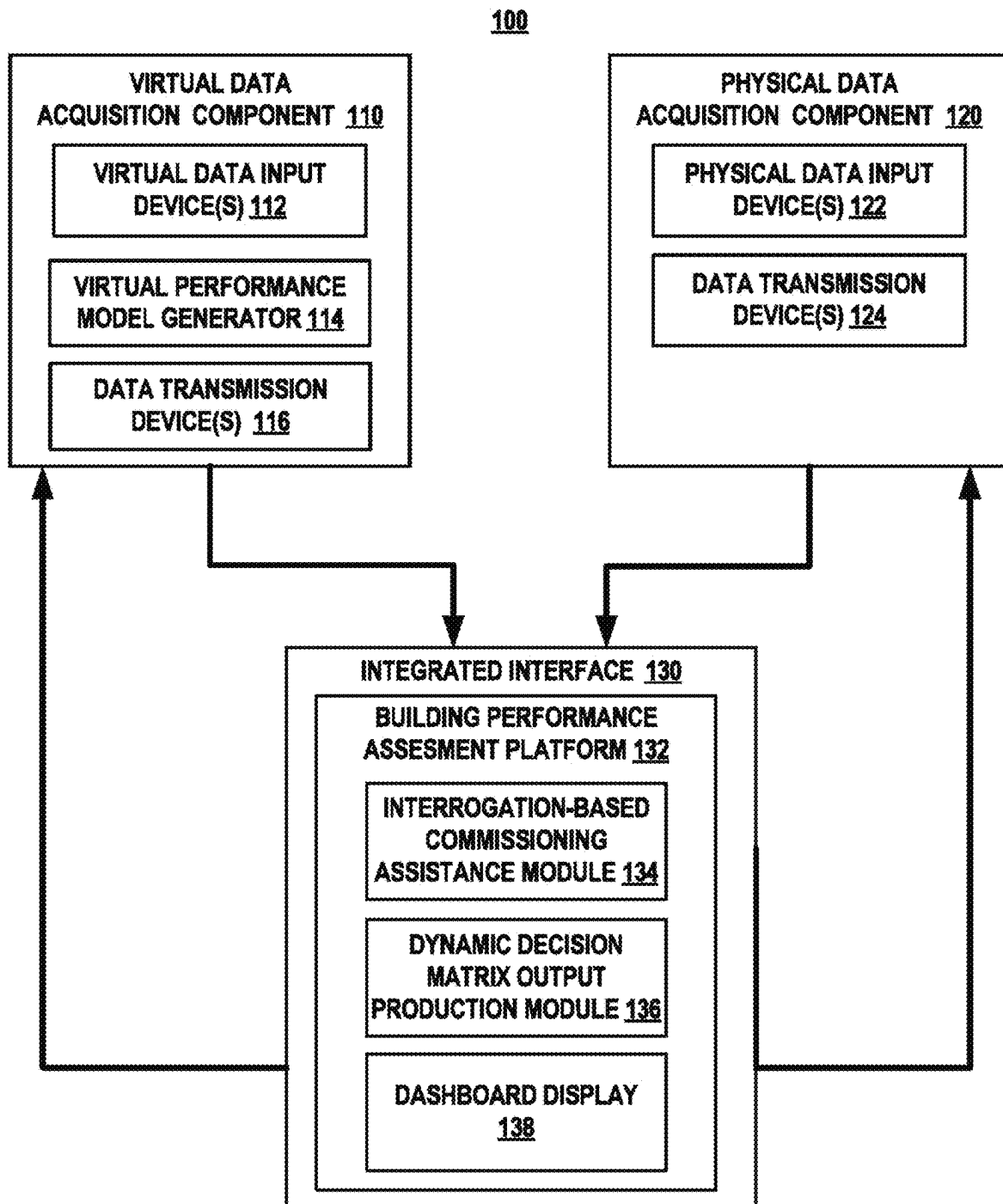
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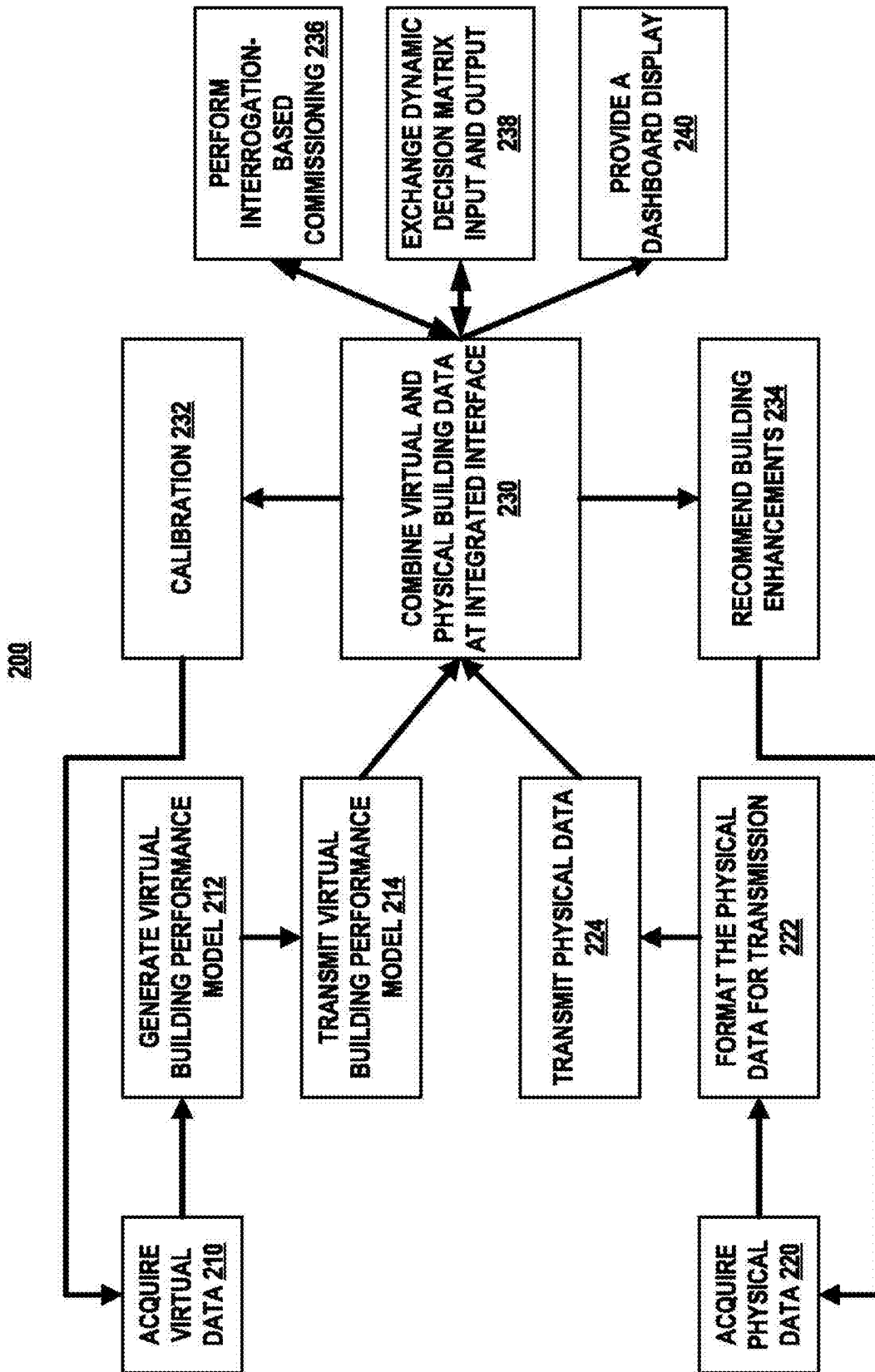
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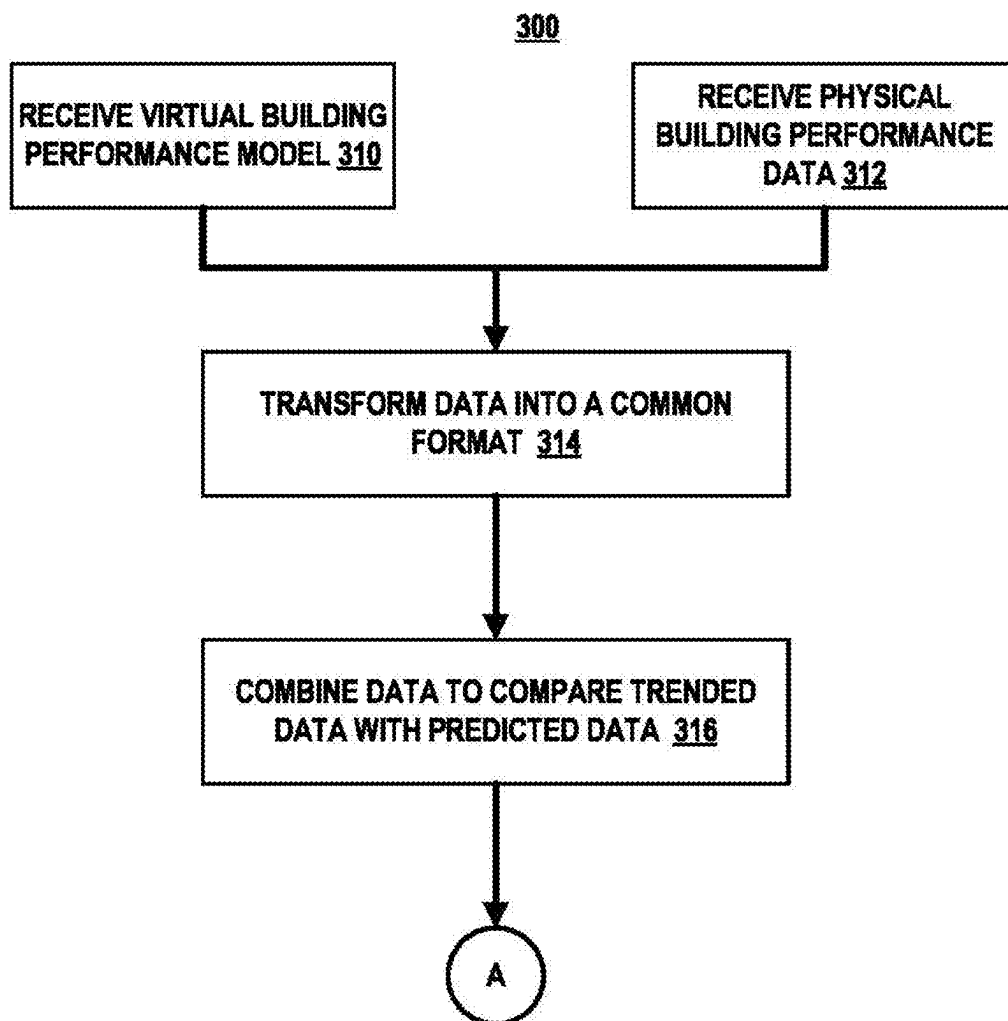
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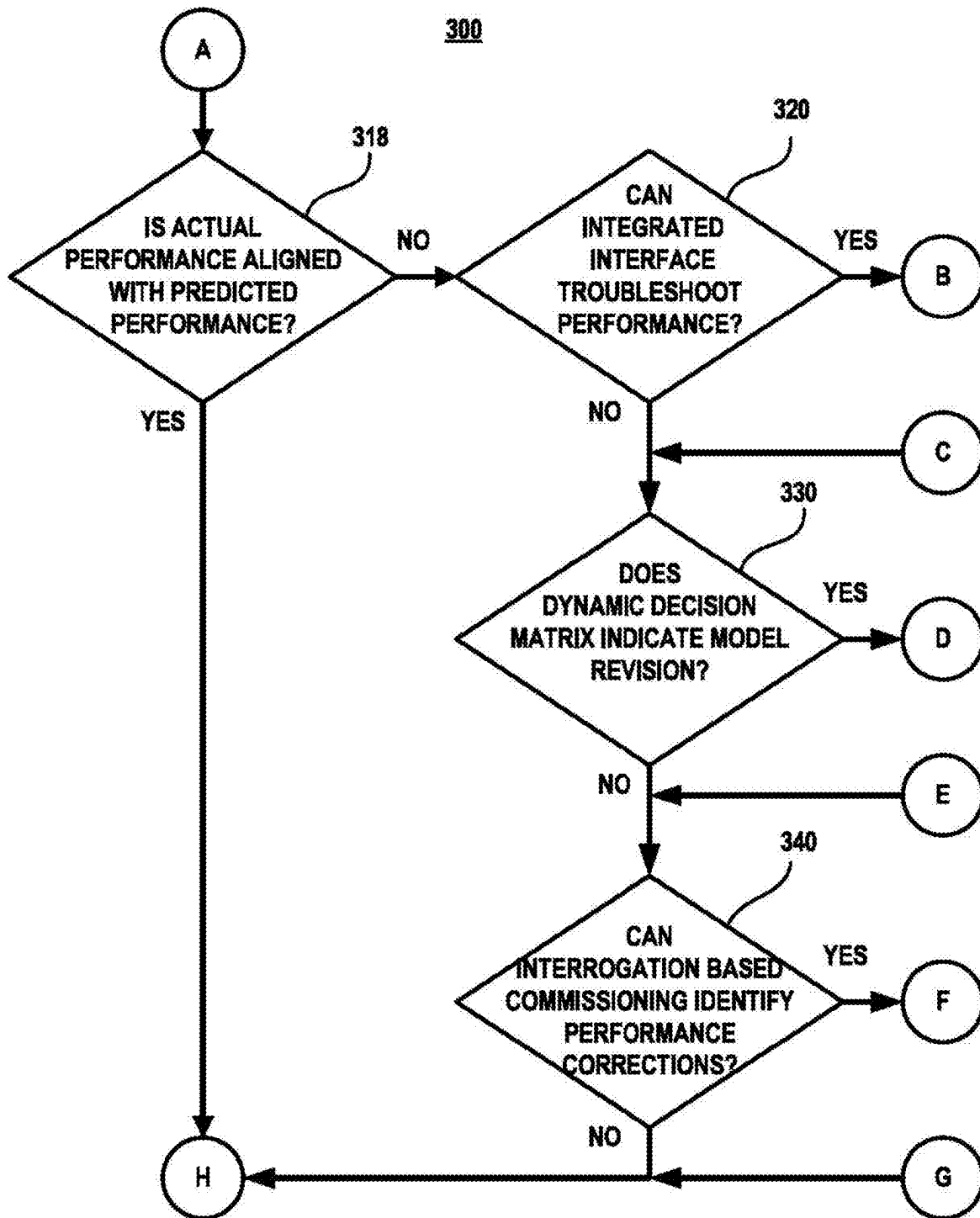
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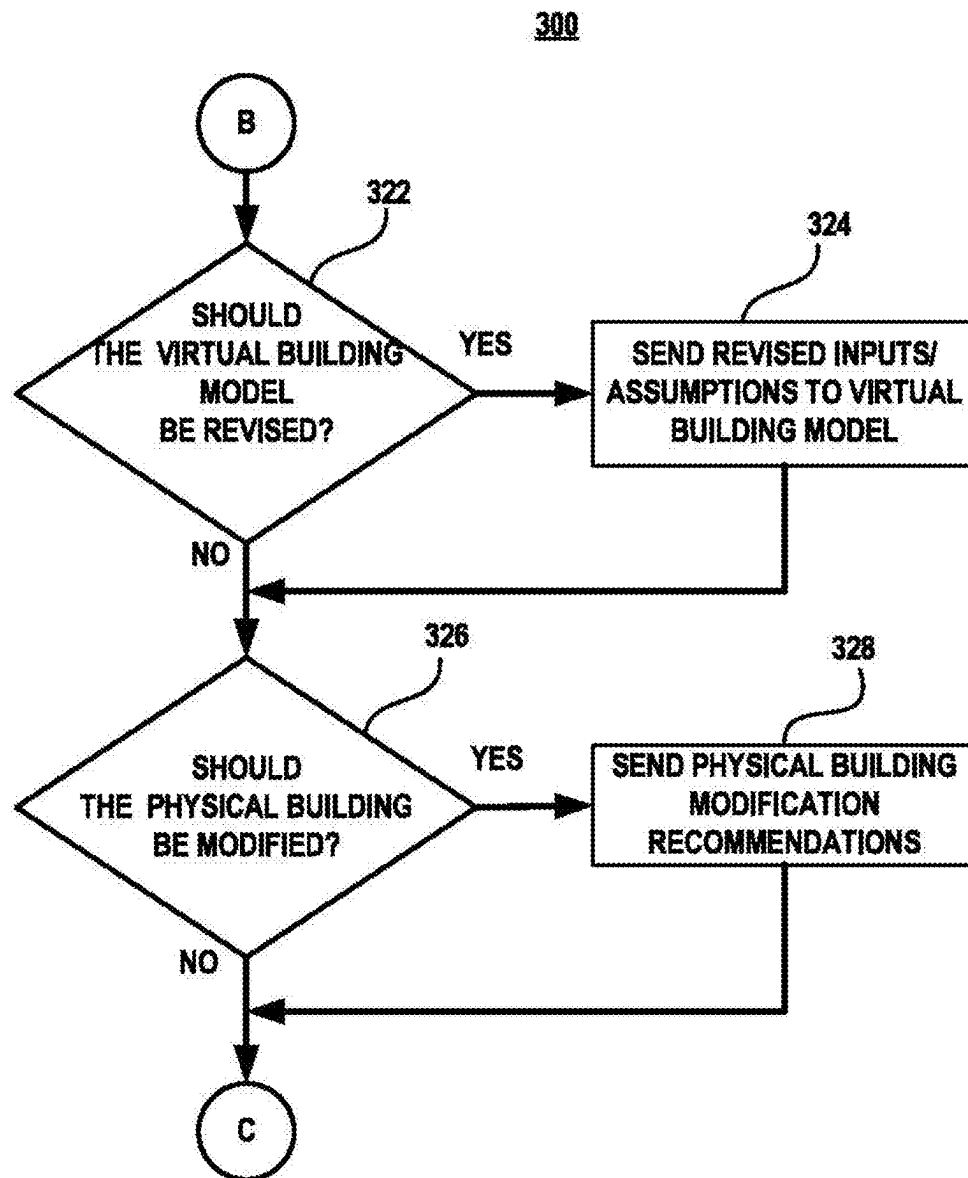
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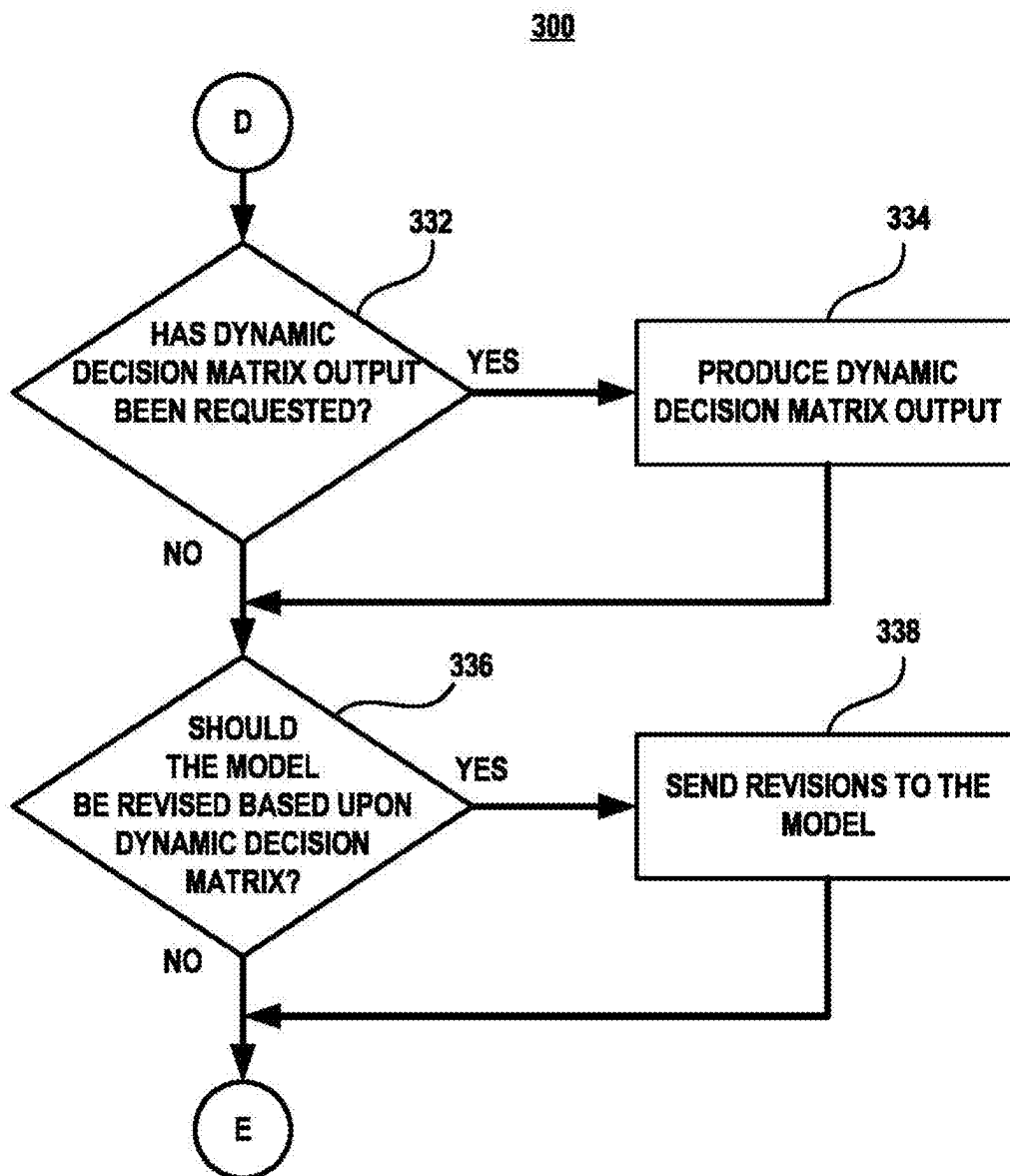
**FIG. 1**

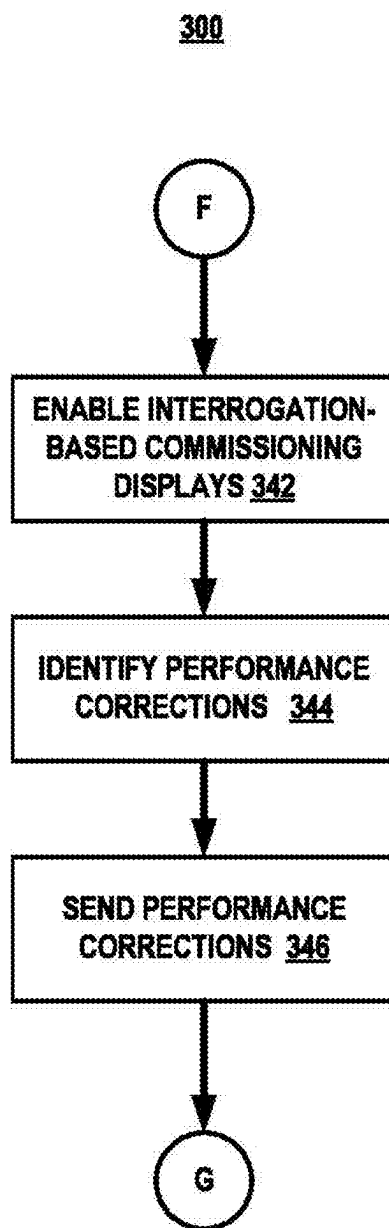
**FIG. 2**

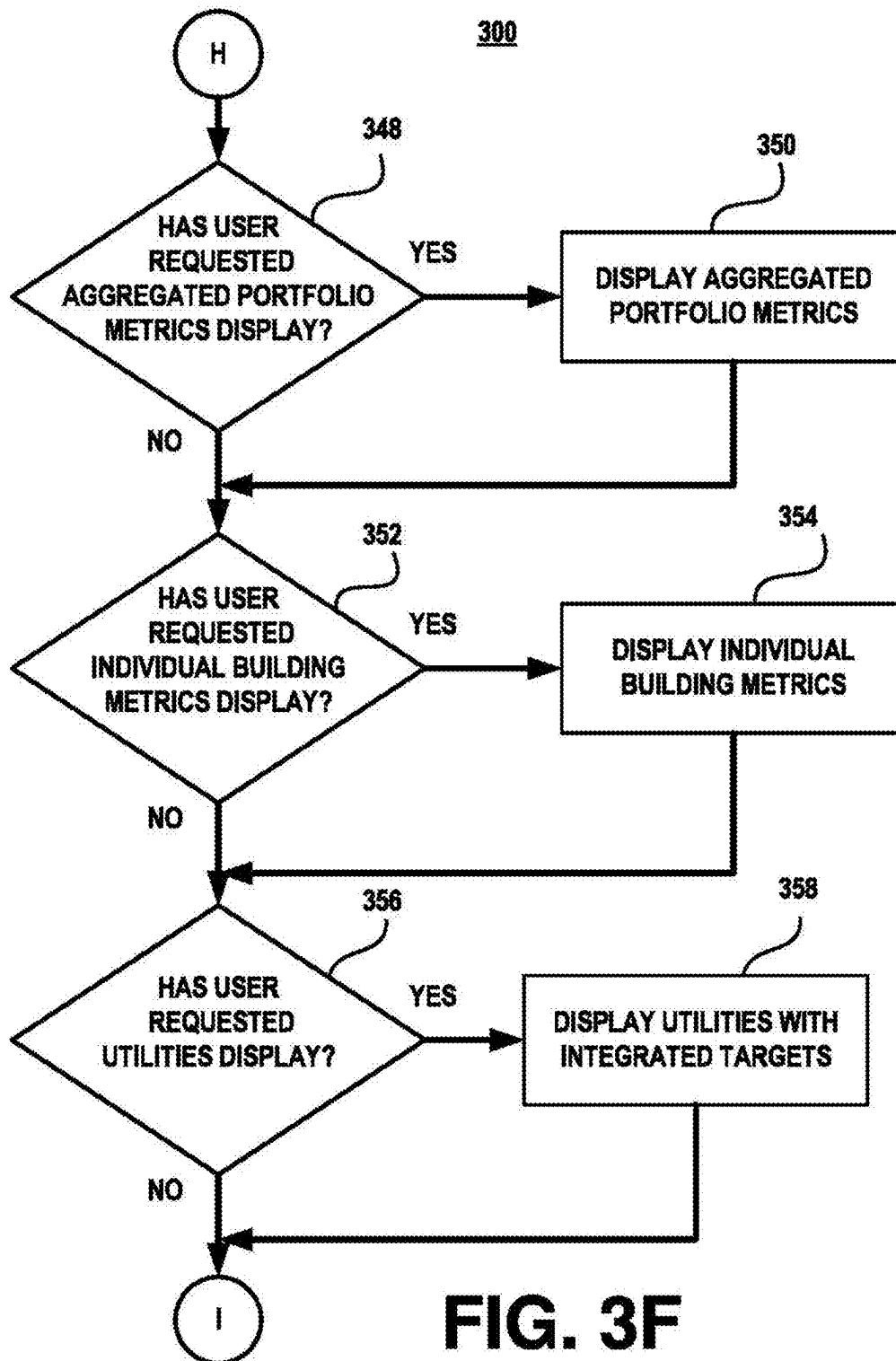
**FIG. 3A**

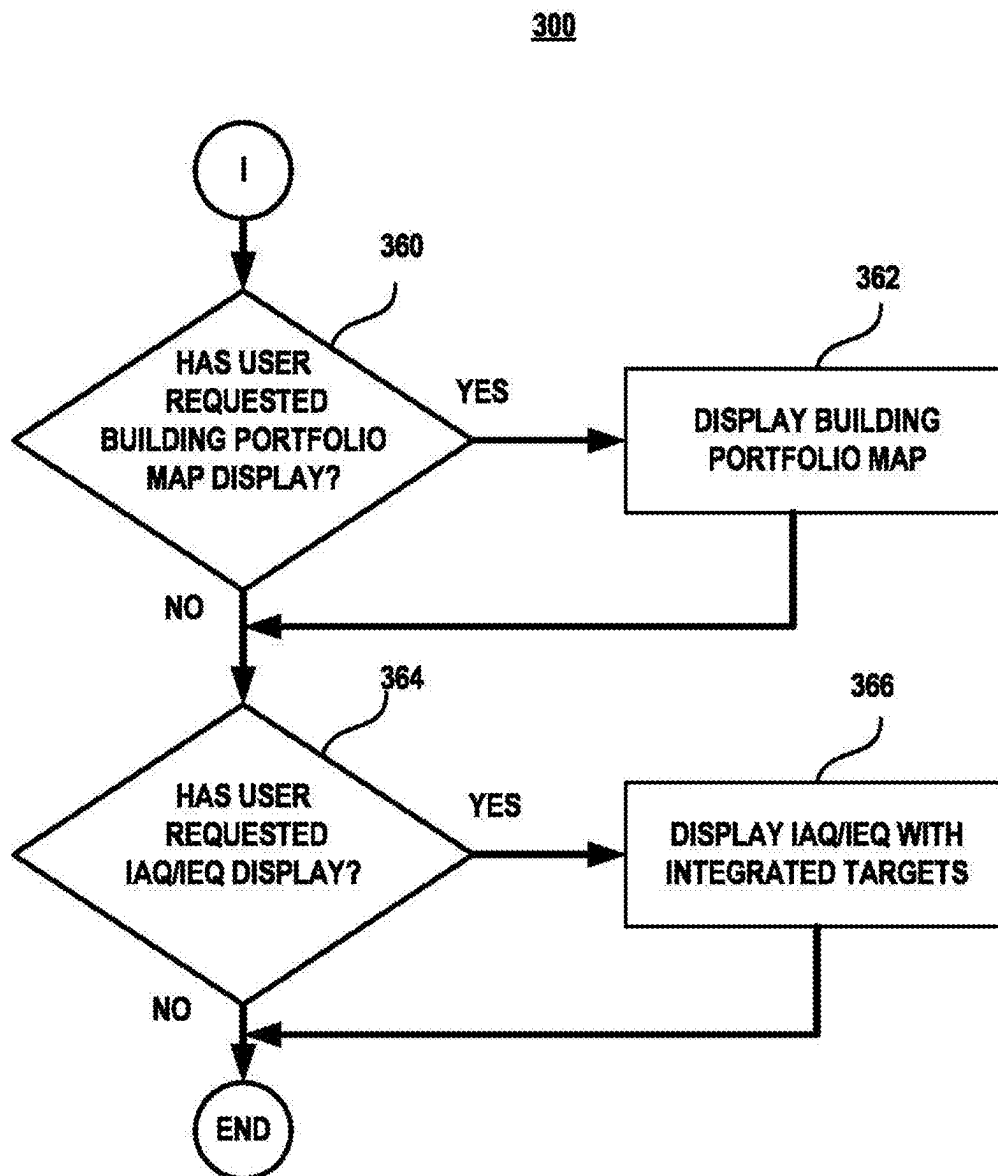
**FIG. 3B**

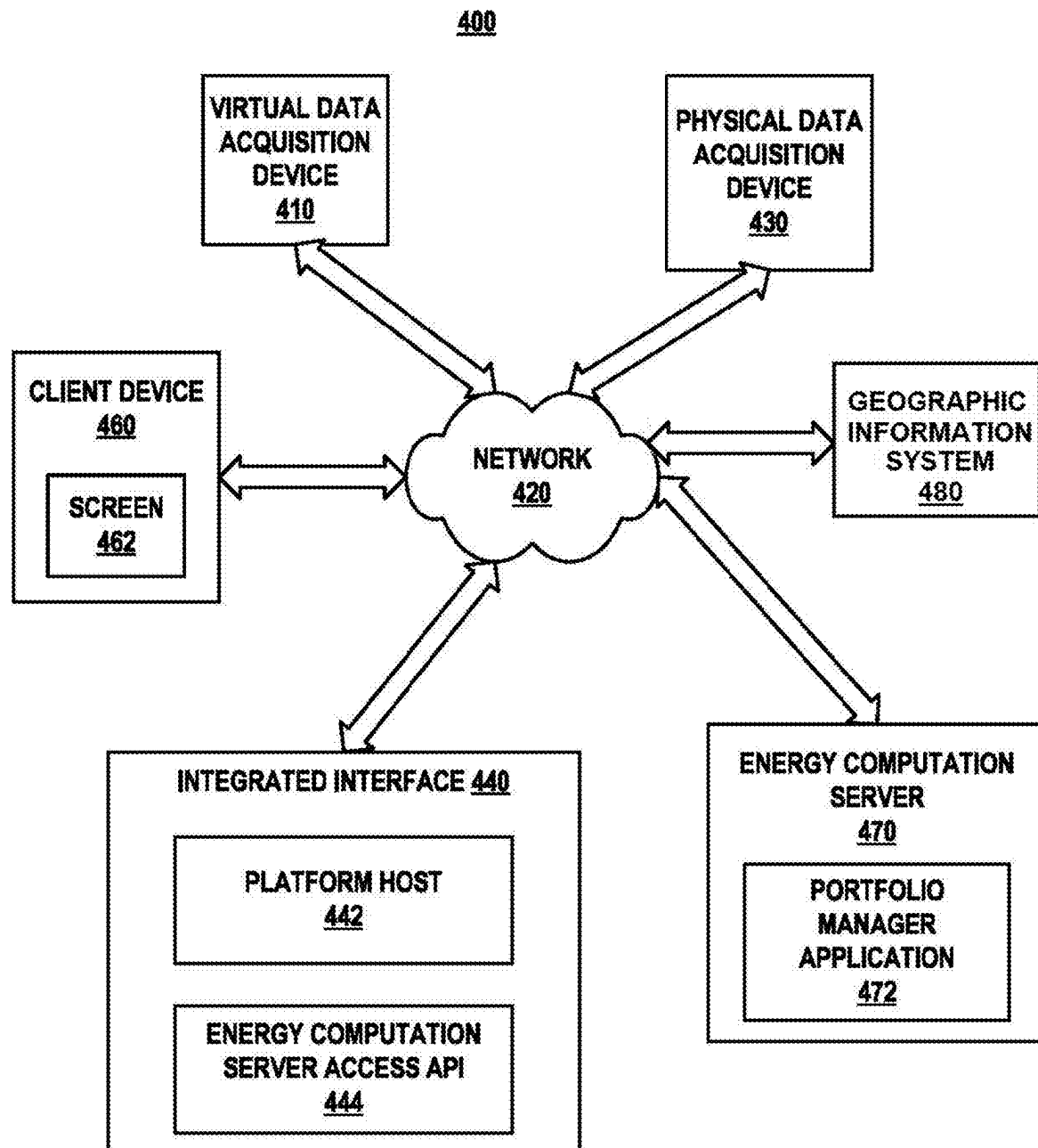
**FIG. 3C**

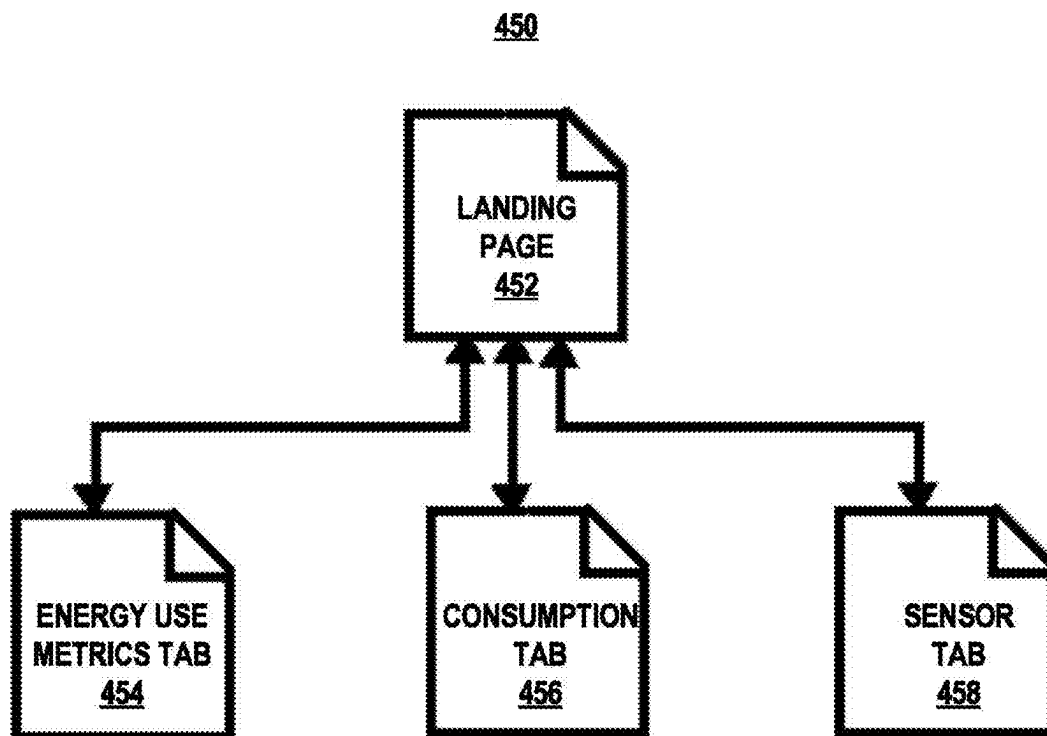
**FIG. 3D**

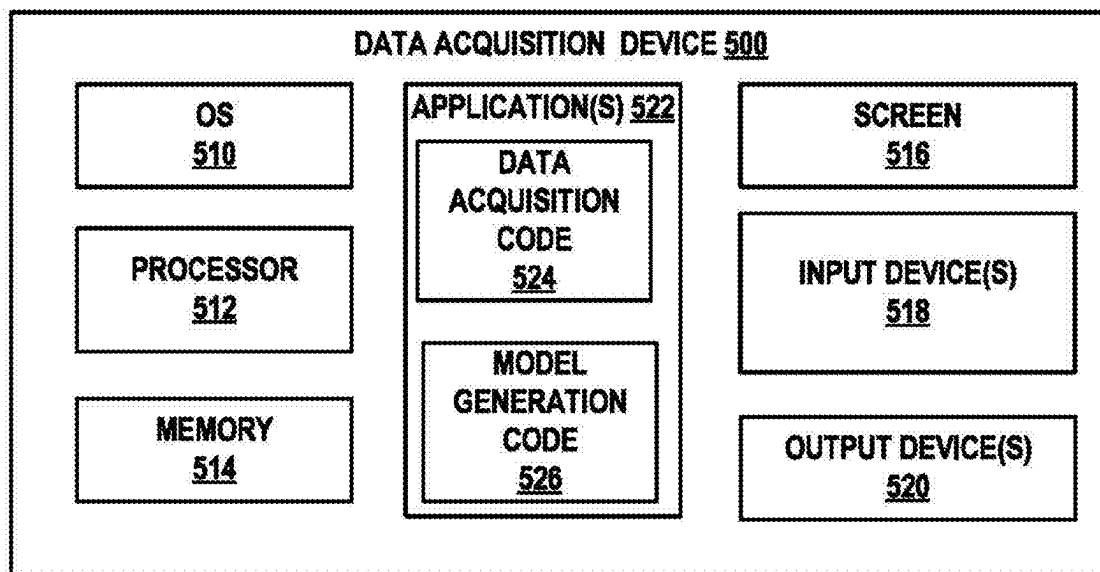
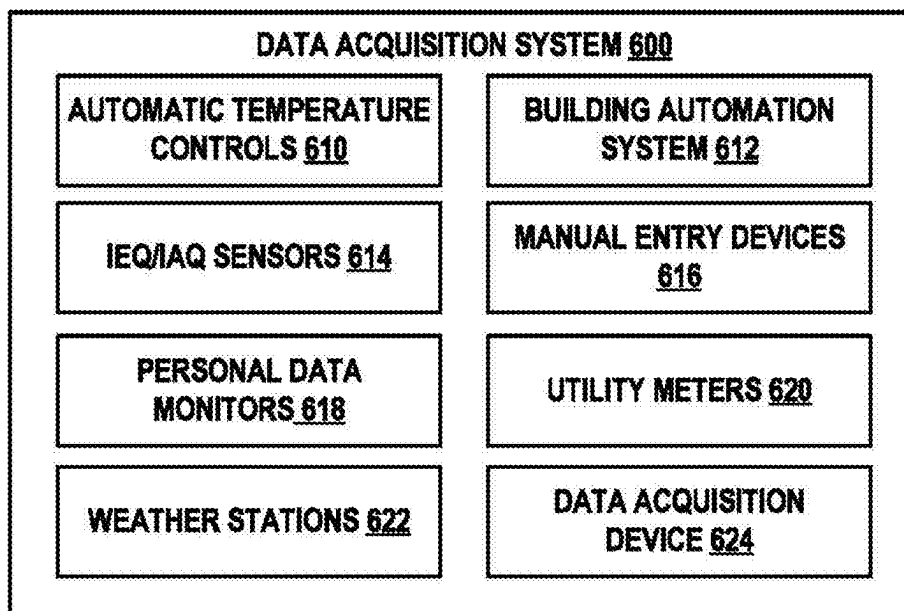
**FIG. 3E**

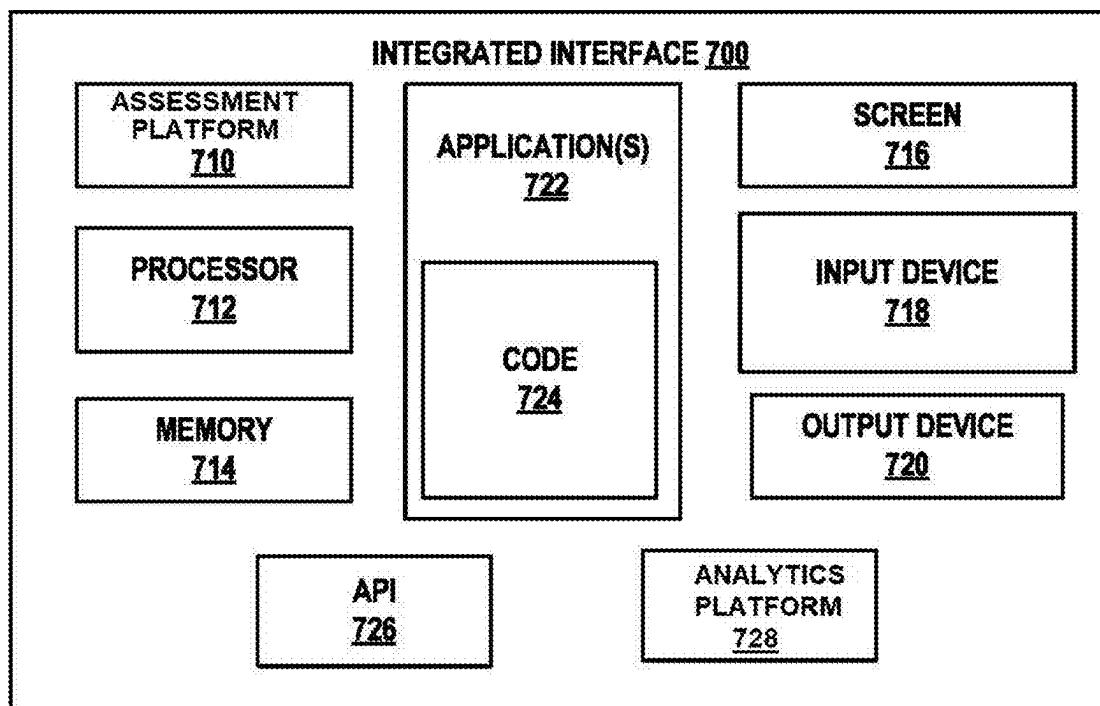
**FIG. 3F**

**FIG. 3G**

**FIG. 4A**

**FIG. 4B**

**FIG. 5****FIG. 6**

**FIG. 7**

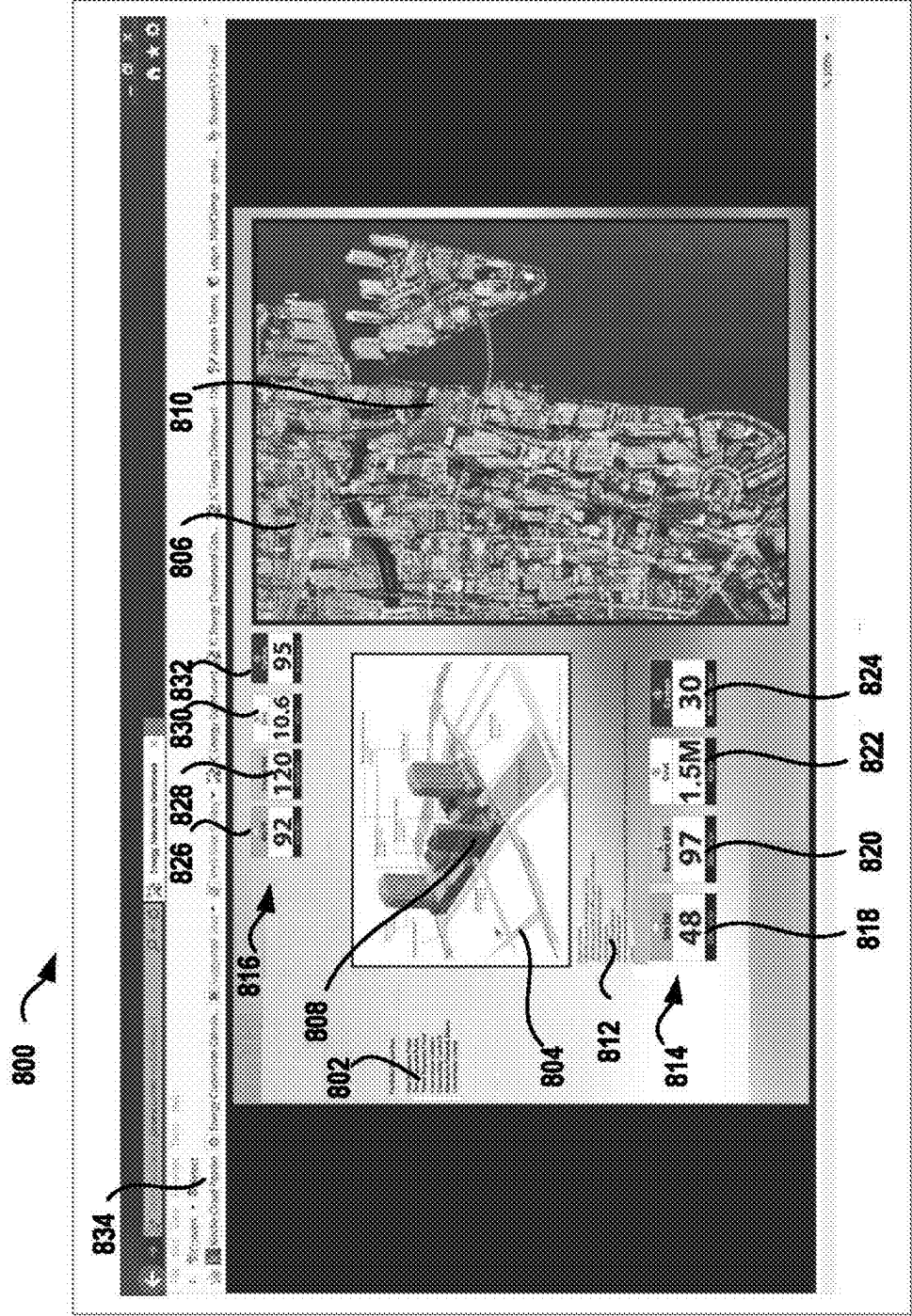


FIG. 8A

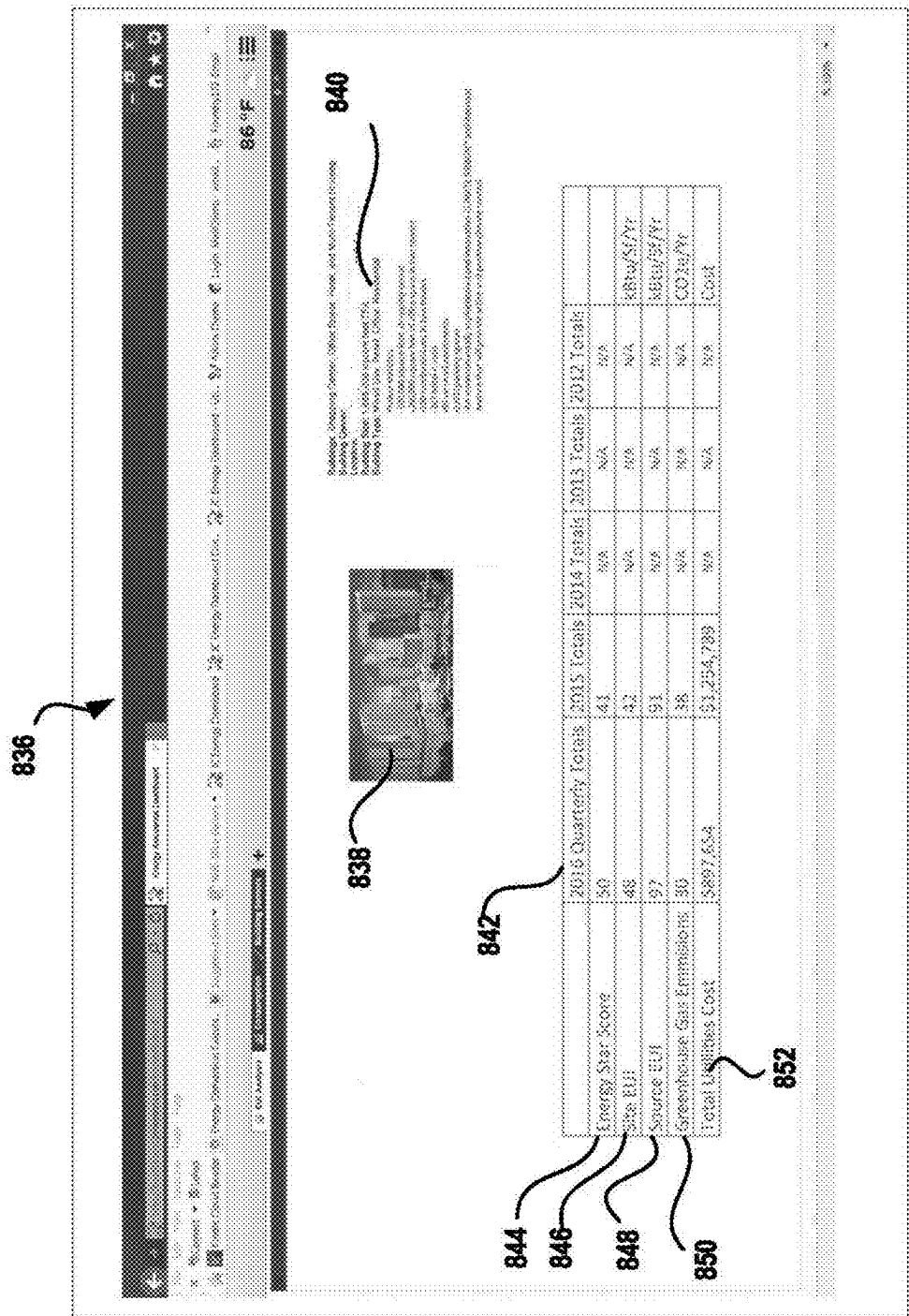


FIG. 8B

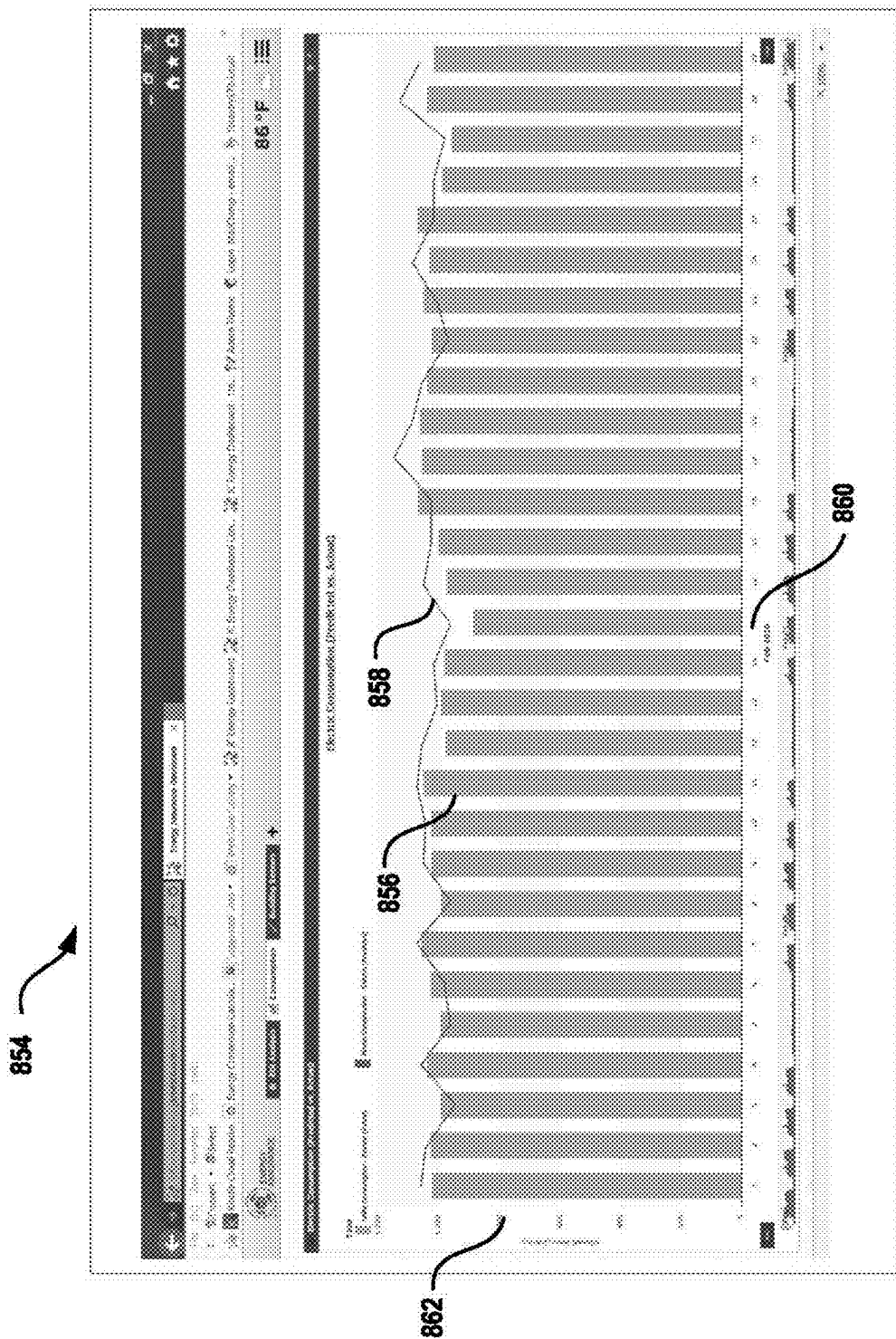


FIG. 8C

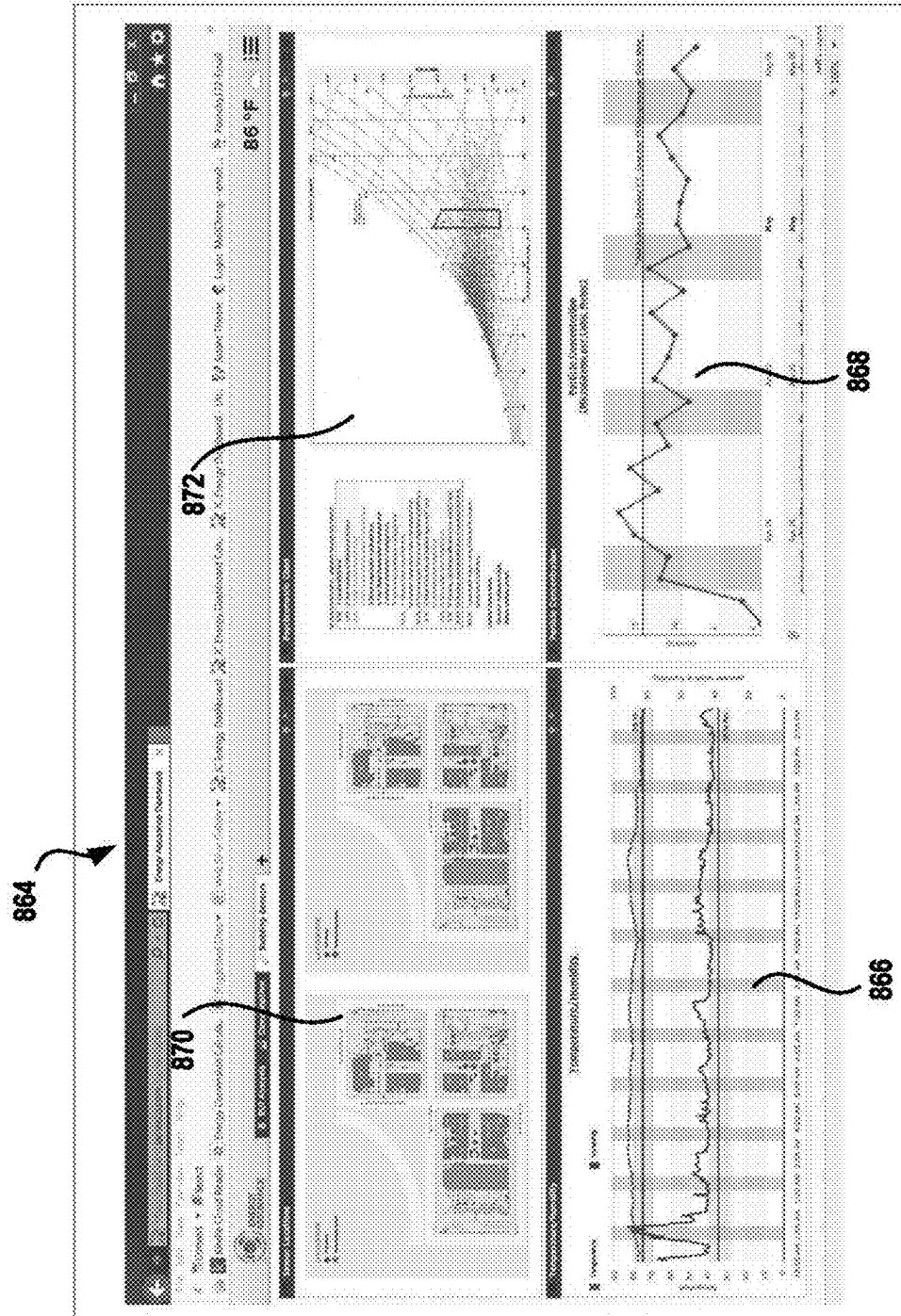
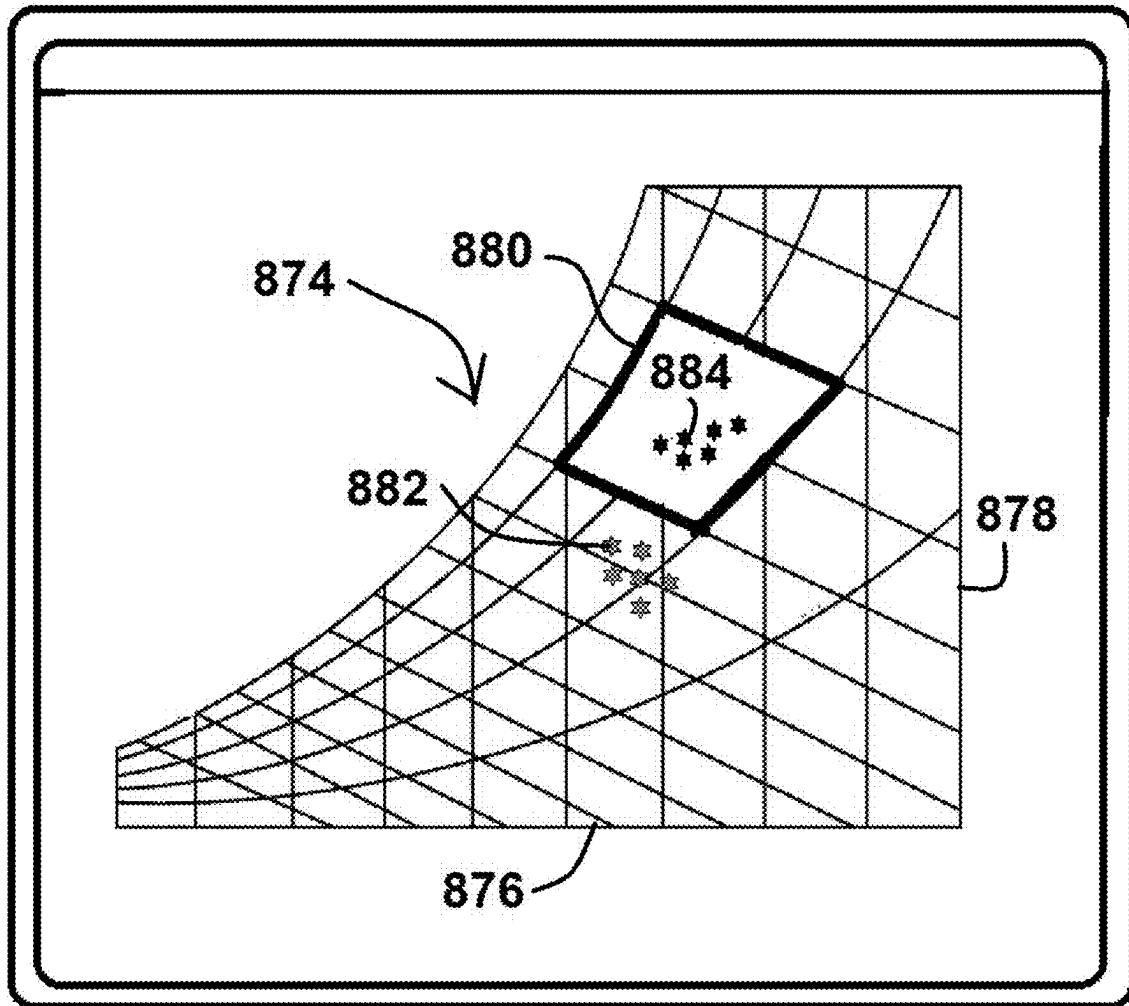
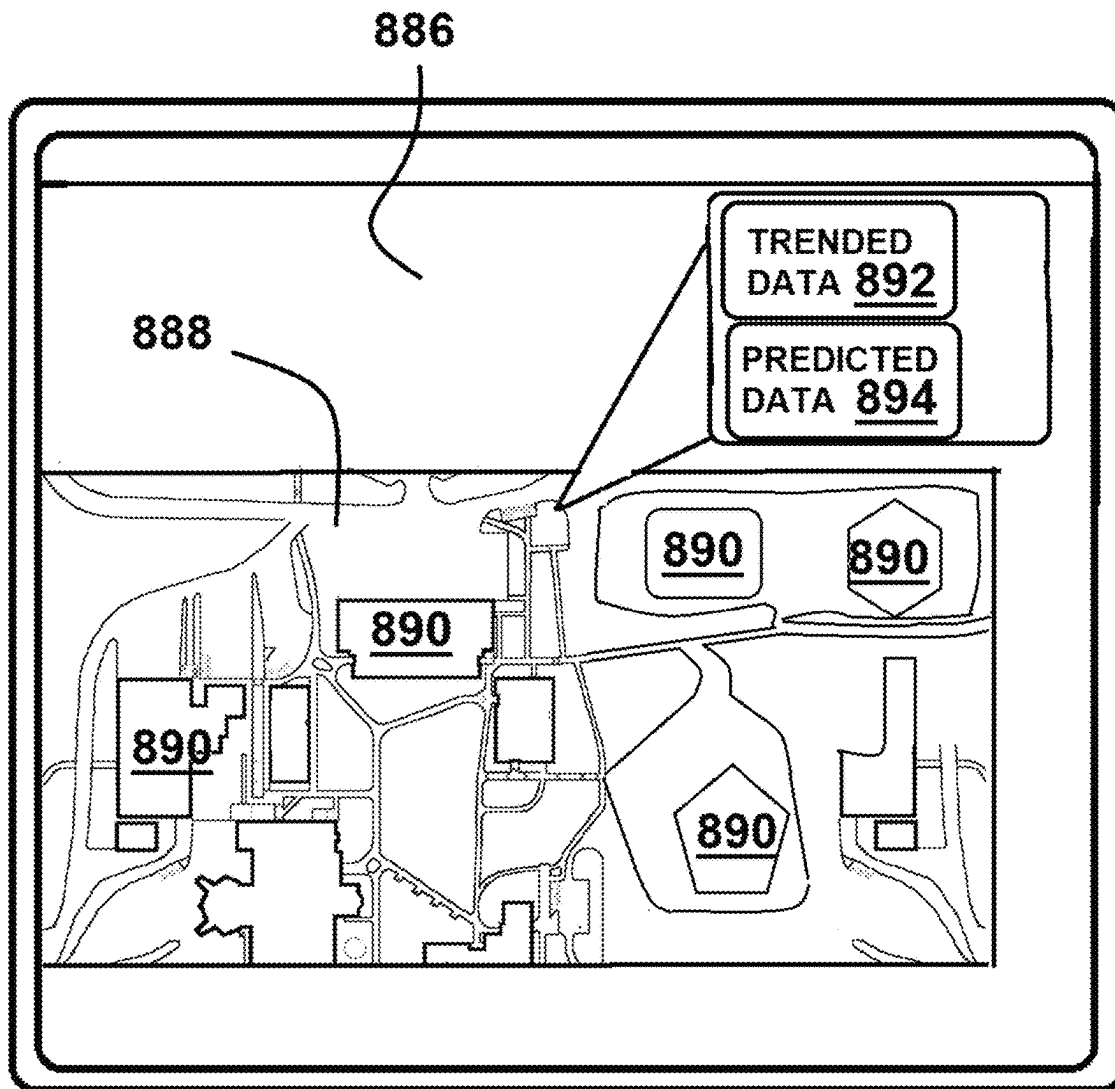
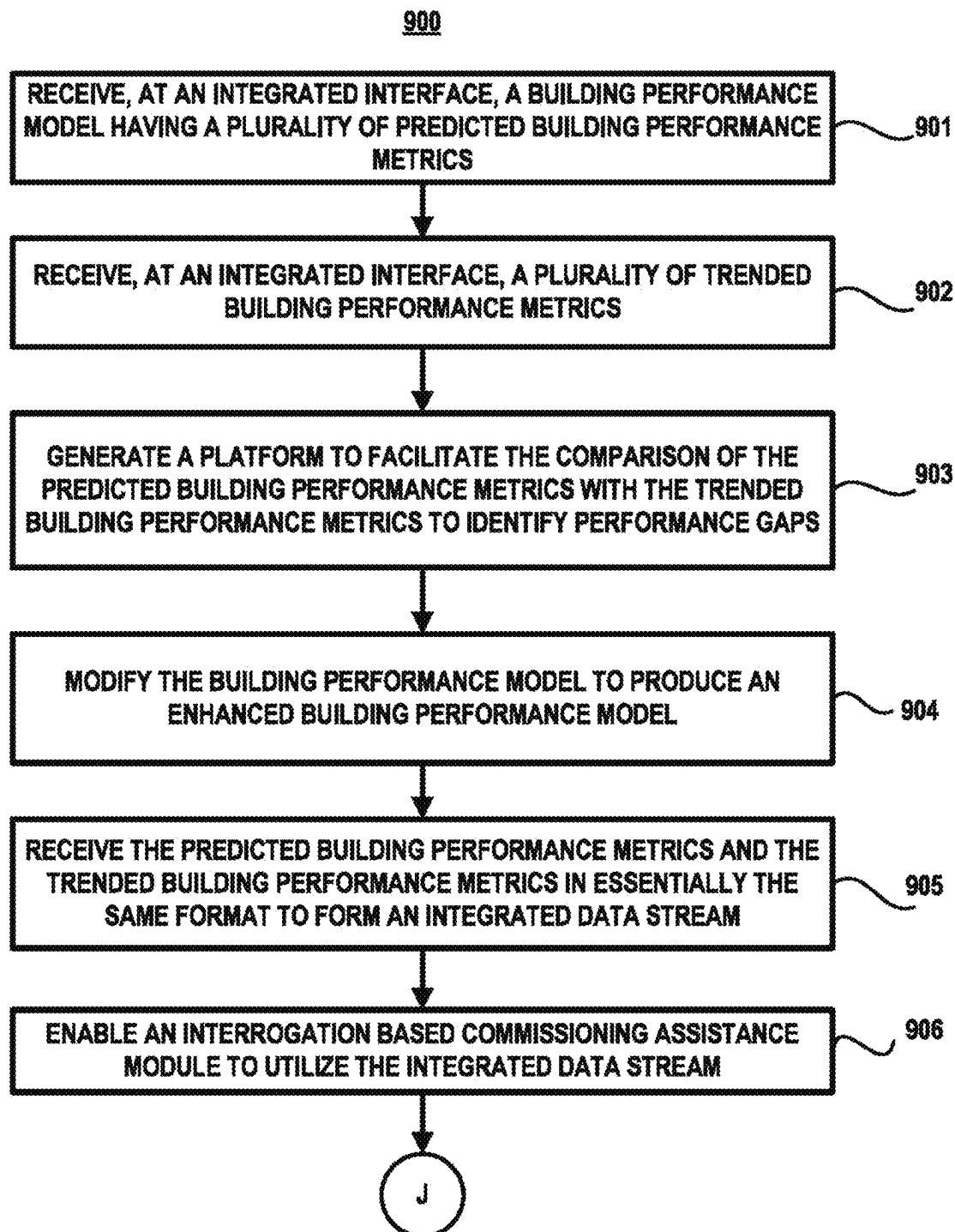
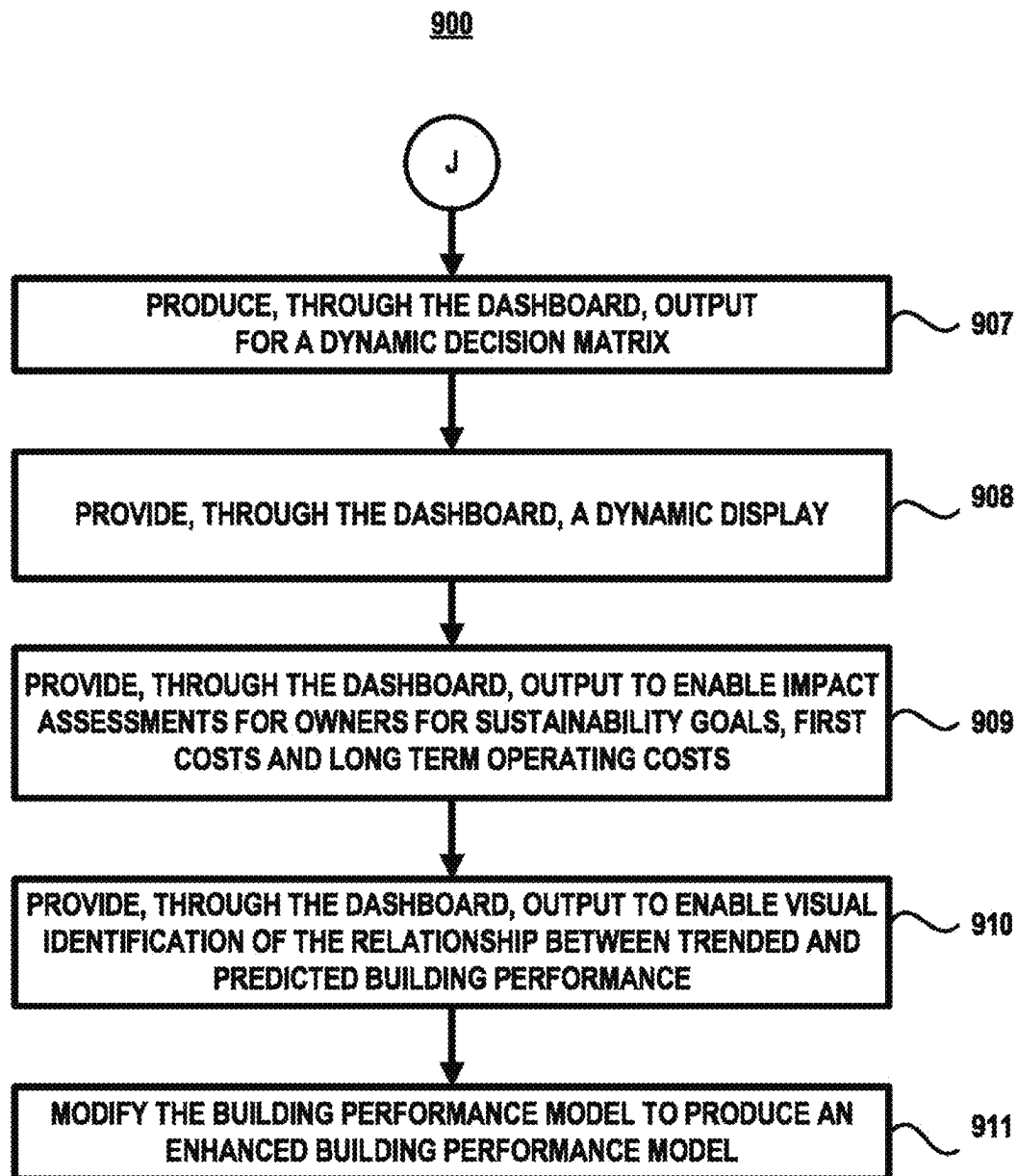


FIG. 8D

**FIG. 8E**

**FIG. 8F**

**FIG. 9A**

**FIG. 9B**

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**BUILDING PERFORMANCE ASSESSMENT
SYSTEM AND METHOD****CROSS REFERENCE TO RELATED
APPLICATIONS**

The present application is a continuation-in-part of U.S. patent application Ser. No. 17/155,222 entitled "BUILDING PERFORMANCE ASSESSMENT SYSTEM AND METHOD" filed Jan. 22, 2021, which is a continuation of U.S. patent application Ser. No. 15/907,222 entitled "BUILDING PERFORMANCE ASSESSMENT SYSTEM AND METHOD" filed Feb. 27, 2018, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 62/530,563 entitled "BUILDING PERFORMANCE ASSESSMENT SYSTEM AND METHOD" filed Jul. 10, 2017, which is incorporated herein by reference.

BACKGROUND

The traditional measures of building performance are limited, primarily, to the measurement of energy usage through energy management systems and/or building automation systems. Energy usage can be a large expense in large commercial facilities that include many buildings. In such facilities, energy usage can cost millions of dollars per year. The challenge in managing these facilities is to minimize the costs and to maintain traditional building performance at acceptable levels.

Building designers, owners, and operators must consider other economic and environmental factors, including Indoor Air Quality (IAQ). IAQ refers to the air quality within and around buildings and structures. In particular, an analysis of IAQ can be used to assess indoor air pollution and the potential health risks that such pollution poses for building occupants. The assessment of IAQ can help building owners understand and control common indoor pollutants to reduce building occupant health concerns.

More recently, the need to assess economic and environmental factors has expanded beyond the assessment of energy usage and IAQ to include an analysis of Indoor Environmental Quality (IEQ). IEQ relates to environmental conditions inside of a building. Assessing IEQ can include evaluating air quality, lighting, views, acoustic conditions, and many other factors. Building managers and operators can increase the satisfaction of building occupants by considering all of the aspects of IEQ rather than narrowly focusing on temperature or air quality alone.

The development of sophisticated computerized energy management systems has provided buildings owners with a variety of new tools that can be used to analyze various aspects of facility operations. These systems can monitor energy usage over time, but they typically compare traditional metrics of buildings performance to historical measurements of building performance, essentially measuring the building against itself. Additionally, these metrics are essentially limited to Site Energy Utilization Intensity (EUI), Source Energy Utilization Intensity (EUI), Cost of Utilities and Greenhouse Gas Emissions (CO₂).

Moreover, these traditional metrics are measured annually and only after a building is operational. This is undesirable because EUI is determined after the building has been operational for some period of time, usually considered to be twelve months. By that time, construction stakeholders have typically exited the project. Since building owners, both for profit and not for profit, cannot afford to wait for twelve months to acquire the trended data that is necessary to obtain

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a base level context for traditional building performance, these existing energy management systems, when used to assess building performance against design, are inadequate.

SUMMARY

The following summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

In various implementations, a system for assessing building performance of a building is provided. A virtual data acquisition device, a physical data acquisition device, an integrated interface having an analytics platform thereon, a client device, and a network connecting the virtual data acquisition device to the physical data acquisition device, the integrated interface, and the client device are provided. The virtual data acquisition device holds a building energy model having a plurality of predicted building performance metrics for the building with the building energy model being a design model. The physical data acquisition device obtains a plurality of trended building performance metrics for the building to transform the building energy model from a design model into an operational model. The integrated interface receives the plurality of predicted building performance metrics from the virtual data acquisition component and the plurality of trended building performance metrics from the physical data acquisition component, wherein the integrated interface analytics platform converts at least one of the plurality of predicted building performance metrics and the plurality of trended building performance metrics into an analytic data product and forms an integrated data stream that includes the analytic data product, and wherein the integrated interface communicates the integrated data stream to a client device for display to provide a user with the ability to compare the analytic data product with at least one of the plurality of predicted building performance metrics and the trended building performance metrics and to identify the building performance gaps therein.

These and other features and advantages will be apparent from a reading of the following detailed description and a review of the appended drawings. It is to be understood that the foregoing summary, the following detailed description and the appended drawings are explanatory only and are not restrictive of various features as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a block diagram of a system for assessing building performance in accordance with the described subject matter.

FIG. 2 illustrates a flow diagram for assessing building performance in accordance with the described subject matter.

FIGS. 3A-3G illustrate a flow diagram for operating the integrated interface shown in FIG. 1 in accordance with the described subject matter.

FIG. 4A illustrates a schematic diagram for another embodiment of a building performance assessment system in accordance with the described subject matter.

FIG. 4B illustrates a schematic diagram for a building performance assessment dashboard in accordance with the described subject matter.

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FIG. 5 illustrates a schematic diagram for a data acquisition device in accordance with the described subject matter.

FIG. 6 illustrates a schematic diagram for a data acquisition system in accordance with the described subject matter.

FIG. 7 illustrates a schematic diagram for an integrated interface in accordance with the described subject matter.

FIG. 8A illustrates a screen shot of a landing page for a dashboard in accordance with the described subject matter.

FIG. 8B illustrates a screen shot of a building metrics page for a dashboard in accordance with the described subject matter.

FIG. 8C illustrates a screen shot of a consumption tab for a dashboard in accordance with the described subject matter.

FIG. 8D illustrates a screen shot of a building sensors tab for a dashboard in accordance with the described subject matter.

FIG. 8E illustrates a screen shot of a psychometric chart in accordance with the described subject matter.

FIG. 8F illustrates a screen shot of a map for a portfolio of buildings in accordance with the described subject matter.

FIGS. 9A-9B illustrate an embodiment of an exemplary process in accordance with the described subject matter.

DETAILED DESCRIPTION

The detailed description provided below in connection with the appended drawings is intended as a description of examples and is not intended to represent the only forms in which the present examples can be constructed or utilized. The description sets forth functions of the examples and sequences of steps for constructing and operating the examples. However, the same or equivalent functions and sequences can be accomplished by different examples.

References to “one embodiment,” “an embodiment,” “an example embodiment,” “one implementation,” “an implementation,” “one example,” “an example” and the like, indicate that the described embodiment, implementation or example can include a particular feature, structure or characteristic, but every embodiment, implementation or example can not necessarily include the particular feature, structure or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment, implementation or example. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, implementation or example, it is to be appreciated that such feature, structure or characteristic can be implemented in connection with other embodiments, implementations or examples whether or not explicitly described.

Numerous specific details are set forth in order to provide a thorough understanding of one or more features of the described subject matter. It is to be appreciated, however, that such features can be practiced without these specific details. While certain components are shown in block diagram form to describe one or more features, it is to be understood that functionality performed by a single component can be performed by multiple components. Similarly, a single component can be configured to perform functionality described as being performed by multiple components.

Buildings consume over 50% of the US energy and have no performance standards that are typically considered during the planning or design phase of construction. Sustainability programs in the built environment are typically prescriptive processes designed to raise awareness. Only recently have sustainability programs started defining acceptable thresholds or ranges of performance for certain

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types of metrics. However, these sustainability programs provide limited, if any, direction on how to cost effectively achieve the performance-oriented requirements, making the adoption of these new programs difficult to scale.

Existing energy management systems do not allow building owners to assess actual building performance compared to project goals set during design due to the fragmented nature of commercial construction. Moreover, such systems are not sophisticated enough to measure a full-range of building performance factors as they relate to project goals. For these reasons there is a need for a holistic system for setting performance goals during planning and measuring performance against those goals in real-time during operations.

Referring to FIG. 1, various features of the subject disclosure are now described in more detail with respect to an integrated building performance assessment system, generally designated as 100. The system 100 addresses the above-described long felt need and allows building owners to assess building performance against project goals established during planning. The system 100 assesses building performance within a more expanded context, such that the relevant building performance metrics include metrics relating to energy usage, IAQ and IEQ. The system 100 defines high performance buildings as any building, group of buildings, sites and/or communities whose performance achieves or exceeds performance targets set during pre-planning, planning, conceptual planning, design, etc.

The system 100 relates, generally, to a system that provides a holistic approach to evidence-based performance in the built environment. The system 100 provides building owners and facility managers with the tools and the data necessary to assess, dynamically, building performance versus the targets set in planning and/or design phases of a project. Stakeholders can use the system 100 throughout the building cycle in a manner that is aligned to deliver common goals, as set in planning.

The system 100 can include a computer system that can configure and implement a virtual data acquisition component 110 that includes one or more virtual data input devices 112, a virtual performance model generator 114, and one or more data transmission devices 116. Additionally, the system 100 can configure and implement a physical data acquisition component 120 that includes one or more physical data input devices 122 and one or more data transmission devices 124. The virtual data acquisition component 110 and the physical data acquisition component 120 can connect to an integrated interface 130.

The integrated interface 130 has the ability to assess environmental impact of energy and indoor environmental quality performance for a building or a group of buildings. The integrated interface 130 measures actual performance in real-time as a function of the financial and social goals of the building owner. These goals can be determined during a planning phase or at another time thereafter.

The integrated interface 130 has the ability to display, visually, actual building performance data versus the goals set during the construction planning process. The integrated interface 130 utilizes hardware, software and proprietary knowledge to show building performance within the context of an investment thesis for a building owner. The integrated interface 130 utilizes whole building energy models and building goals that can be developed during the planning process. These models can be revisited or tracked, routinely, once a building is in operation. In this exemplary embodiment, the models can predict various indicators of building

performance, including metrics for site EUI, source EUI, cost, carbon emissions and/or other greenhouse gas emissions.

The integrated interface **130** provides building owners with transparency with respect to the delivery of expected building performance during the construction process. The integrated interface **130** can link building planning to building operations. The integrated interface **130** provides building owners with the ability to set performance goals during the design and planning stages of construction and to track, digitally, actual building performance against those goals for troubleshooting purposes.

The integrated interface **130** can include a computer system or computing device that configures and implements a building performance assessment platform **132**. The building performance assessment platform **132** implements and utilizes an interrogation-based commissioning assistance module **134**, a dynamic decision matrix output production module **136**, and a dashboard display **138**.

The virtual data acquisition component **110** implements and utilizes the virtual data input devices **112** to obtain information relating to various building performance metrics. The information can be obtained from architects, designers, construction managers, modelers, engineers, and sustainability consultants. The information can relate to architectural design, cost estimates, simulated occupancy, subcomponent energy models, weather station output, whole-building energy models and/or other inputs. The information can be obtained manually or through automated data acquisition methods or systems.

The virtual performance model generator **114** can utilize the above-described information to generate a virtual building performance model having a plurality of predicted building performance metrics. The virtual building performance model can include predicted biometrics, predicted IAQ/IEQ, and/or predicted utility consumption. The predicted utility consumption can be categorized by sources, including cogeneration systems, district energy chilled-water, district energy steam, electric, gas, potable water, renewable geothermal, renewable photovoltaics, renewable wind, and/or one or more other types of utility. The virtual building performance model can be a design model or an operational model depending upon when it is generated or modified.

The virtual data acquisition component **110** can transmit the virtual performance model to the integrated interface **130** using the data transmission devices **116**. The virtual performance model can be transmitted via an application programming interface (API), a cloud computing system, email, geofencing, a hardwired network, a proprietary device driver, a WiFi network, a wireless network, a Bluetooth network, and/or through any other suitable transmission method, device, or network. In some embodiments, the data transmission devices **116** can connect to the integrated interface **130** directly, so that data that is collected from the data input devices **112** is transmitted to the integrated interface **130** directly.

The physical data acquisition component **120** can configure and implement the physical data input devices **122** to collect physical data. The physical data can relate to sensor output and/or data output relating to ambient lighting, biometrics, carbon dioxide (CO₂), carbon monoxide (CO), circadian lighting, formaldehyde, humidity, internally generated noise, lead (Pb), ozone (O₃), particulate matter (PM_{2.5} & PM₁₀), radon (Rn), sound pressure level, temperature, volatile organic compounds (VOCs), water quality and/or other data types relating to energy usage or indoor

environmental quality. The physical data input devices **122** can obtain data manually or through automated data acquisition methods or systems.

The physical data input devices **122** can include automatic temperature controls, building automation system, design and engineering, IAQ/IEQ sensors, manual entry devices, personal data monitors, utility meters, weather stations and/or other inputs, as available.

The physical data acquisition component **120** can communicate the information from the physical data input devices **122** to the data transmission devices **124**. The data transmission devices **124** can use the information to generate or to calculate a plurality of trended building performance metrics. The trended building performance metrics can include trended biometrics, trended IAQ/IEQ, and trended utility. The physical building performance metrics can be transmitted to the building performance assessment platform **132** on the integrated interface **130**.

The physical data acquisition component **120** can transmit the physical building performance metrics and/or the raw data to the building performance assessment platform **132** on the integrated interface **130** using the data transmission devices **124**. The information can be transmitted via an application programming interface (API), a cloud computing system, email, geofencing, a hardwired network, a proprietary device driver, a WiFi network, a wireless network, a Bluetooth network, and/or through any other suitable transmission method, device, or network. In some embodiments, the data transmission devices **124** can connect to the integrated interface **130** directly, so that data that is collected from the data input devices **122** is transmitted to the building performance assessment platform **132** on the integrated interface **130** directly.

The virtual data acquisition component **110** can generate models during various construction phases, including planning phases, design phases, calibration phases, and operation phases. The physical data acquisition component **120** can obtain the physical building performance metrics during a calibration phase. The integrated interface **130** can obtain the physical building performance metrics from the physical data acquisition component **120** during the calibration phase to transform the model from a design model into an operational model using trended data to inform the design model. In some embodiments, the planning phases and the design phases can last one year, the calibration phases can last one year, and the operational phase can last one year.

The integrated interface **130** can receive the predicted building performance metrics and the trended building performance metrics in essentially the same format to form an integrated data stream. The integrated interface **130** can enable the modification of the virtual building performance model to produce an enhanced building performance model. A builder can use the integrated data stream to modify its building plans or the building, itself. The integrated data stream can be a data stream in which the data is in the same format (e.g., unit of measure and timescale) and displayed in a manner that allows comparison.

The integrated interface **130** can convert the physical building performance metrics, the raw data, and/or the predicted building performance metrics into analytics. For example, certain metrics can be utilized to determine analytics that indicate carbon consumption and/or production by a building.

The integrated interface **130** can configure and implement a building performance assessment platform **132**. The building performance assessment platform **132** can utilize an

interrogation-based commissioning assistance module **134**, a dynamic decision matrix output production module **136**, and a dashboard display **138**.

The interrogation-based commissioning assistance module **134** can utilize the integrated data stream to modify the whole-building performance model based upon comparisons of trended data with predicted data. The interrogation-based commissioning assistance module **134** can use the trended data and/or the predicted data to perform physics-based simulations that can be used to identify and to remediate problems, which can improve whole building performance. The simulations can use new building profiles and set new targets to produce operation-based solutions.

The dynamic decision matrix output production module **136** can utilize the integrated data stream to provide output for a dynamic decision matrix. The dynamic decision matrix can include information relating to building performance goals, construction costs and long term operating costs. The output can be used by a building owner to compare multiple solution sets for improving building performance.

The dynamic decision matrix can be the product of financial modeling and/or a financial calculator. The financial calculator can be an energy use intensity (EUI) and IEQ value financial calculator. The calculator can provide the ability prove tangible financial savings that can be achieved through investment into energy conservation and improved IEQ measures. The financial models produced by the calculator can predict the potential cost or benefit of EUI and IEQ performance in whole building environments.

The calculator uses connectivity to calculate the value provided to building owners of new, existing and retrofit commercial and residential buildings. The calculator can identify the value associated with operational benefit, the social cost of carbon benefits, asset valuation benefits, and/or occupant benefits.

The calculator can be a tool for predicting and validating the savings associated with EUI and IEQ improvements as a function of investment costs. The calculators can provide a holistic view of return on investment for energy conservation and IEQ related investments.

The building performance assessment platform **132** can utilize the dashboard display **138** to provide output for visually assessing a plurality of key performance indicators for a building. The dashboard display **138** can provide output to enable visual identification of the relationship between trended and predicted building performance, which can enable the comparison of predicted building performance metrics with trended building performance metrics to identify gaps between predicted building performance and actual building performance. The dashboard display **138** can be used to identify these gaps at any time during the entire building lifespan, including the planning phases, the initial construction phases, the operation phases, and, if applicable, the building renovation phases, for new buildings, existing buildings, and/or retrofit buildings.

Referring to FIG. 2 with continuing reference to the foregoing figures, an exemplary process, generally designated by the numeral **200**, for operating an integrated building performance assessment system is shown. In this exemplary embodiment, the process **200** can be performed by the integrated building performance assessment system **100** shown in FIG. 1.

At **210**, virtual data is acquired. In this exemplary embodiment, virtual data is acquired using the virtual data acquisition component **110** shown in FIG. 1. The virtual data acquisition component **110** uses the virtual data input devices **112** to acquire the virtual data. The virtual data is

communicated to the virtual performance model generator **114**, which generates a virtual building performance model at **212**. The virtual building performance model is transmitted to the integrated interface **130** shown in FIG. 1 at **214**.

At **220**, physical data is acquired. In this exemplary embodiment, physical data is acquired using the physical data acquisition component **120** shown in FIG. 1. The physical data acquisition component **120** uses the physical data input devices **122** to acquire the physical data. The physical data is communicated to the data transmission device **124** for formatting at **222**. The physical building performance data is transmitted to the integrated interface **130** shown in FIG. 1 at **224**.

At **230**, the integrated interface combines the data from the virtual building performance model with the physical building performance data into an integrated data stream. In this exemplary embodiment, the integrated interface can be the integrated interface **130** shown in FIG. 1. The integrated interface **130** can use the data to calibrate the virtual building performance at **232** and use the data to recommend physical building enhancements (i.e., to drive building performance) at **234**. The calibration at step **232** transforms the virtual building performance model from a design model into an operational model.

The integrated data stream can be divided into two categories of data. The first data category can include utility data types. Utility data types include cogeneration system data, district energy chilled-water data, district energy steam data, electric utility data, gas utility data, water utility data, various renewable energy data and/or any other type of relevant utility data. The second data category can include physical data relating to IAQ/IEQ, which includes ambient lighting, biometrics, CO₂, CO, circadian lighting, formaldehyde, humidity, internally generated noise, Pb, O₃, PM2.5, PM10, Rn, sound pressure level, temperature, VOCs, water quality and/or other relevant, related data, as described above.

At **236**, interrogation-based commissioning can be performed. In this exemplary embodiment, interrogation-based commissioning can be performed by the interrogation-based commissioning assistance module **134** shown in FIG. 1.

At **238**, dynamic decision matrix output and input can be exchanged. In this exemplary embodiment, dynamic decision matrix output and input can be exchanged by the dynamic decision matrix output production module **136** shown in FIG. 1. The dynamic decision matrix output can include data relating to building performance goals, building performance costs, and long term operating costs.

At **240**, a dashboard display can be provided. In this exemplary embodiment, the integrated interface **130** can implement the dashboard display **138** shown in FIG. 1. The dashboard display **138** can display, dynamically, aggregate portfolio metrics, individual building metrics, utility metrics, IAQ/IEQ metrics, and/or building portfolio maps and/or metrics.

Referring to FIGS. 3A-3G with continuing reference to the foregoing figures, an exemplary process, generally designated by the numeral **300**, for operating an integrated interface is shown. In this exemplary embodiment, the process **300** can be performed by the integrated interface **130** shown in FIG. 1.

At **310**, the integrated interface **130** receives a virtual building performance model. In this exemplary embodiment, the integrated interface **130** receives the virtual building performance model from the virtual data acquisition component **110** shown in FIG. 1. The virtual building

performance model having a plurality of predicted metrics, including energy consumption, IAQ and/or IEQ metrics.

At **312**, the integrated interface **130** receives physical building performance metrics and/or data. In this exemplary embodiment, the integrated interface **130** receives the metrics and/or data from the physical data acquisition component **120** shown in FIG. 1. The metrics and/or data can include a plurality of trended or measured metrics, including energy consumption, IAQ and/or IEQ metrics.

At **314**, the integrated interface **130** can transform the data into a common format to form an integrated data stream. At **316**, the data can be combined to compare trended data with predicted data.

At **318**, the integrated interface **130** can be utilized to determine whether actual performance aligns with predicted performance. If the integrated interface **130** determines that actual performance is not aligned with predicted performance, then the integrated interface **130** can be utilized to determine whether the actual and/or the predicted performance can be improved through troubleshooting at **320**.

At **322**, the integrated interface **130** determines whether the virtual building performance model should be revised. If the integrated interface **130** determines that the model should be revised, then the integrated interface **130** communicates with the virtual data acquisition component **110** to revise the model at **324**. The virtual data acquisition component **110** can revise the model by recalibrating the virtual data input devices **112** or by adjusting the virtual performance model generator **114** shown in FIG. 1.

At **326**, the integrated interface **130** determines whether the physical building should be modified. If the integrated interface **130** determines that the physical building should be modified, then the integrated interface **130** sends building modification recommendations at **328**. The building modification recommendations can be displayed by the display device **138** shown in FIG. 1.

At **330**, the integrated interface **130** configures and implements the dynamic decision matrix output production module **136** shown in FIG. 1 to determine whether the model should be revised because of performance gaps. In this exemplary embodiment, the integrated interface **130** determines whether dynamic decision matrix output has been requested at **332**. If the output has been requested, the integrated interface **130** produces dynamic decision matrix output.

The dynamic decision matrix can be used to determine the impact of an enhanced building performance model on various building performance goals and/or costs, which can be based upon actual building performance. The goals and/or costs can include sustainability goals, first costs, and long term operating costs.

The dynamic decision matrix output can be used to determine whether the model should be revised at **336**. If the integrated interface **130** determines that the model should be revised, the integrated interface **130** can communicate with the virtual data acquisition component **110** to revise the model at **338**. It should be understood that the dynamic decision matrix output can be used to determine whether the building should be reconfigured, retrofitted, and/or renovated.

At **340**, the integrated interface **130** configures and implements the interrogation based-commissioning assistance module **134** shown in FIG. 1 to determine whether the models should be revised because gaps exist. In this exemplary embodiment, the integrated interface **130** enables interrogation-based commissioning displays at **342**. The displays can produce output relating to area personnel use,

building automation systems, design and engineering modifications, electrical systems, envelop performance, field changes, indoor environmental quality, mechanical systems, outside climate, quality control testing, thermal comfort, value-engineering and/or any other relevant type of output.

The interrogation-based commissioning assistance module **134** can interact with building owners and/or other users to identify corrections at **344**. The performance corrections can be sent through the integrated interface **130** to the virtual data acquisition component **110**, to the physical data acquisition component **120**, or to the dashboard display **138** at **346**.

The integrated interface **130** can configure and implement the dashboard display **138** shown in FIG. 1. The operation of the dashboard display **138** is represented by steps **348-366** in FIGS. 3F-3G.

At **348**, the integrated interface **130** determines whether a user has requested an aggregated portfolio metrics display. If the output has been requested, the dashboard display **138** generates an aggregated portfolio metrics display at **350**. The aggregated portfolio metrics can include EnergyStar® portfolio manager output, site EUI, source EUI, carbon, and/or costs. EnergyStar® is a registered trademark of the Environmental Protection Agency.

At **352**, the integrated interface **130** determines whether a user has requested an individual building metrics display. If the output has been requested, the dashboard display **138** generates an individual building metrics display at **354**. The individual building metrics can include EnergyStar® portfolio manager output, site EUI, source EUI, carbon, and/or costs.

At **356**, the integrated interface **130** determines whether a user has requested a display of utility information with integrated targets thereon. If the output has been requested, the dashboard display **138** generates the display at **358**. The utility information can include metrics relating to cogeneration systems, district energy chilled-water, district energy steam, electric, gas, potable water, renewable geothermal, renewable photovoltaics, renewable wind, or any other type of utility.

At **360**, the integrated interface **130** determines whether a user has requested a building portfolio map. If the output has been requested, the dashboard display **138** generates a building portfolio map at **362**. The building portfolio map can include one or more building high performance indicators.

At **364**, the integrated interface **130** determines whether a user has requested an IAQ/IEQ display with integrated targets. If the output has been requested, the dashboard display **138** generates an IAQ/IEQ display with integrated targets at **366**. The display can include metrics and/or targets relating to ambient lighting, biometrics, CO₂, CO, circadian lighting, formaldehyde, humidity, internally generated noise, Pb, O₃, PM_{2.5}, PM₁₀, Rn, sound pressure level, temperature, VOCs, water quality and/or other relevant, related data, as described above.

Referring now to FIGS. 4A-4B with continuing reference to the foregoing figures, another embodiment of a building performance assessment system, generally designated as **400**, is illustrated. It is to be appreciated that sections, components, and/or portions of the building performance assessment system **400** can be implemented as a computer, computer system, and/or computing device and can be configured as a special purpose computer or a general purpose computer specifically programmed to generate building performance models. Further, it is to be appreciated these features can be implemented by various types of

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operating environments, computer networks, platforms, frameworks, computer architectures, and/or computing devices.

Implementations of a computer system are described within the context of a system configured to perform various steps, methods, and/or functionality in accordance with the described subject matter. It is to be appreciated that a computer system can be implemented by one or more computing devices. Implementations of the computer system can be described in the context of “computer-executable instructions” that are executed to perform various steps, methods, and/or functionality in accordance with the described subject matter.

In general, computers, computer systems, and/or computing devices can include one or more processors and storage devices (e.g., memory and disk drives) as well as various input devices, output devices, communication interfaces, and/or other types of devices. Such systems and devices can include a combination of hardware and software. It can be appreciated that various types of computer-readable storage media can be part of a computer, computer system, and/or computing device. As used herein, the terms “computer-readable storage media” and “computer-readable storage medium” do not mean and unequivocally exclude a propagated signal, a modulated data signal, a carrier wave, or any other type of transitory computer-readable medium. In various implementations, a computer system can include a processor configured to execute computer-executable instructions and a computer-readable storage medium (e.g., memory and/or additional hardware storage) storing computer-executable instructions configured to perform various steps, methods, and/or functionality in accordance with aspects of the described subject matter.

Computer-executable instructions can be embodied and/or implemented in various ways such as by a computer program (e.g., client program and/or server program), a software application (e.g., client application and/or server application), software code, application code, source code, executable files, executable components, routines, application programming interfaces (APIs), functions, methods, objects, properties, data structures, data types, and/or the like. Computer-executable instructions can be stored on one or more computer-readable storage media and can be executed by one or more processors, computing devices, and/or computer systems to perform particular tasks or implement particular data types in accordance with aspects of the described subject matter.

A computer, computer system, and/or computing device can implement and utilize one or more program modules. Generally, program modules include routines, programs, objects, components, data structures, etc., that perform particular tasks or implement particular abstract data types.

A computer, computer system, and/or computing device can be implemented as a distributed computing system or environment in which components are located on different computing devices that are connected to each other through network (e.g., wired and/or wireless) and/or other forms of direct and/or indirect connections. In such distributed computing systems or environments, tasks can be performed by one or more remote processing devices, or within a cloud of one or more devices, that are linked through one or more communications networks. In a distributed computing environment, program modules may be located in both local and remote computer storage media including media storage devices. Still further, the aforementioned instructions may be implemented, in part or in whole, as hardware logic circuits, which may or may not include a processor.

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A computer system can include one or more servers. Servers can be implemented by one or more computing devices such as server computers configured to provide various types of services and/or data stores in accordance with aspects of the described subject matter. Exemplary servers computers can include, without limitation: web servers, front end servers, application servers, database servers, domain controllers, domain name servers, directory servers, and/or other suitable computers.

Components of computers, computer systems, and/or computing devices can be implemented by software, hardware, firmware or a combination thereof. For example, computer systems can include components implemented by computer-executable instructions that are stored on one or more computer-readable storage media and that are executed to perform various steps, methods, and/or functionality in accordance with aspects of the described subject matter.

As shown in FIGS. 4A-4B, the system 400 includes a virtual data acquisition device 410 that has the ability to acquire virtual data or metrics relating to predicted building performance based upon input from one or more devices and/or sources. The virtual data acquisition device 410 can generate a virtual building performance model or digital twin based the virtual data or metrics. The virtual building model can include predicted energy consumption metrics, predicted indicators, predicted measurements, and/or predicted sensor outputs. The models can be whole building energy models that establish building goals during the planning process.

In some embodiments, the digital twin can be an integrated digital twin. An integrated digital twin represents the real-time integration of physics-based simulation and trended data for a building or groups of buildings.

The virtual data acquisition device 410 connects to a network 420, which is connected to a physical data acquisition device 430 that can have a plurality of sensors and/or sensor measurement systems that can be used to acquire trended data and trended sensor measurements.

The network 420 can be implemented by any type of network or combination of networks including, without limitation: a wide area network (WAN) such as the Internet, a local area network (LAN), a Peer-to-Peer (P2P) network, a telephone network, a private network, a public network, a packet network, a circuit-switched network, a wired network, and/or a wireless network. Servers and workstations can communicate via networks using various communication protocols (e.g., Internet communication protocols, WAN communication protocols, LAN communications protocols, P2P protocols, telephony protocols, and/or other network communication protocols), various authentication protocols, and/or various data types (web-based data types, audio data types, video data types, image data types, messaging data types, signaling data types, and/or other data types).

The physical data acquisition device 430 has the ability to acquire observed or trended data or metrics relating to actual building performance based upon input from one or more devices and/or sources. The observed or trended data can include observed or trended building performance metrics, trended indicators, trended measurements, and/or trended sensor outputs. The trended metrics, indicators, measurements and/or sensor outputs can be acquired in real-time or near real-time.

The physical data acquisition device 430 can convert raw sensor output into actual building performance metrics to report the metrics over the network 420. Alternatively, the physical data acquisition device 430 can report actual raw

sensor output over the network **420** for conversion into actual building performance metrics by an integrated interface **440**.

The integrated interface **440** connects to the virtual data acquisition device **410** and the physical data acquisition device **430** over the network **420**. The integrated interface **440** includes a platform host **442** and an API **444**. The integrated interface **440** can be configured to implement the platform host **442** to receive the virtual building performance model from the virtual data acquisition device **410** and/or the physical building performance metrics or raw data from the physical data acquisition device **430**. The integrated interface **440** can communicate with the virtual data acquisition device **410** and/or the physical data acquisition device **430** to revise and/or to track the model, the data and/or the metrics, as necessary.

The platform host **442** can generate a platform that enables a dashboard **450** for the comparison of predicted building performance metrics and/or indicators from the virtual building performance model with trended building performance metrics and/or indicators. The dashboard **450** can be an integrated building performance dashboard that includes a graphical display that provides for the comparison of actual building performance to simulated targets set during building design and/or planning. Building performance can be based upon various metrics relating to all sources of energy, IAQ and/or IEQ.

The dashboard **450** can provide users with the ability to compare actual building performance for a building with respect to energy usage, indoor air quality metrics, indoor environmental quality metrics, indicators, measurements, and/or sensor output against performance goals for the building set at the time of investment. As a result, the dashboard **450** provides building owners with the ability to assess return on investment and/or to hold vendors accountable for successful delivery of products or services.

The dashboard **450** can include a landing page **452** and various tabs, such as an energy use metrics tab **454**, a consumption tab **456**, and a sensor tab **458**. The landing page **452** can include at least one building icon relating to a building and links to the energy use metrics tab **454**, the consumption tab **456**, and the sensor tab **458**. The dashboard **450** can facilitate real-time analysis of utility consumption and/or investment targets at a particular point of investment, planning, discovery and/or design.

The integrated interface **440** connects to one or more client device(s) **460** over the network **420**. The client device(s) **460** can include one or more building monitor displays, desktop PCs, laptops, tablets, iPads, iPhones, smartphones or other similar communication devices. The client devices **450** can be enabled to display the landing page **452**, the energy use metrics tab **454**, the consumption tab **456**, and the sensor tab **458**. The client device(s) **460** can include a screen **462**.

The consumption tab **456** can display the at least one trended building performance metric with a corresponding predicted building performance metric, so that the trended building performance metric can be compared to the predicted building performance metric for a time interval within a predetermined period of time.

The sensor tab **458** can display, graphically, sensor output, utility meter output, or other similar output from the physical data acquisition device **430**. The sensor tab **158** can display sensor output in a manner that conveys utility consumption. The output can be used to determine various actual energy usage, environmental quality, and/or other building performance metrics for the energy use metrics tab **454**, the

consumption tab **456**, and/or the sensor tab **458**. The output can be superimposed with target metrics at equivalent time intervals.

IEQ information can be displayed by the dashboard **450** by physical area, sensor type and uses graphical icons to identify individual sensors. The visual display of actual building performance versus simulated targets at time of investment shows provides real-time analysis of building performance.

A building owner can use the information obtained from the dashboard **450** and displayed on the landing page **452**, the energy use metrics tab **454**, the consumption tab **456** and/or the sensor tab **458** to use whole building modeling and the dashboard to troubleshoot, digitally, building performance attributes during the building interrogation process post occupancy. The dashboard **450** can be used to perform interrogation-based commissioning processes that use digital simulation to narrow and to clarify the areas of performance gaps.

The API **444** connects the integrated interface **440** to an energy computation server **470** that hosts an energy portfolio manager application **472**. In this exemplary embodiment, the energy computation server **470** is a server that is hosted by EnergyStar®. The energy portfolio manager application **472** can generate energy use metrics output for display on the energy use metrics tab **454**.

In some embodiments, the integrated interface **440** can communicate through the network **420** with a geographic information system (GIS) that includes software, such as the Aeronautical Reconnaissance Coverage Geographic Information System (ArcGIS), Google Earth, Google Maps, and AutoCAD MAP.

Referring now to FIG. 5 with continuing reference to the foregoing figures, a data acquisition device, generally designated as **500**, is illustrated. The data acquisition device **500** can be implemented as a computer, computer system, and/or computing device that can be configured as a special purpose computer or a general purpose computer specifically programmed to acquire data. The data acquisition device **500** can be configured to implement the virtual data acquisition component **110** shown in FIG. 1 and/or the virtual data acquisition device **410** shown in FIGS. 4A-4B.

The data acquisition device **500** includes an operating system **510**, a processor **512**, memory **514**, a screen **516**, an input device **518**, and an output device **520**. The building model generation device **500** can host one or more applications **522** that include data acquisition code **524** and/or model generation code **526**.

The processor **512** can perform tasks such as signal coding, data processing, input/output processing, power control, and/or other functions. Memory **514** can be used for storing data and/or code for running operating system **510** and/or application(s) **522**. Example data can include web pages, text, images, sound files, video data, or other data to be sent to and/or received from one or more network servers or other devices via one or more wired and/or wireless networks.

The operating system **510**, the processor **512**, memory **514**, and/or application(s) **522** can cooperate to utilize screen **516** and/or to communicate with input device **518** and/or output device **520**.

The application(s) **522** can produce virtual building performance models that predict building performance during the planning phase. The virtual building performance models can be whole-building energy models that can be used to create alternate scenarios for building performance. The models can include utility consumption, IAQ and/or IEQ

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targets that can be provided to the dashboard 450 shown in FIGS. 4A-4B. The targets can be labels that are displayed simultaneously with actual building performance. The targets can be displayed for predetermined scenarios.

The application(s) 522 can create bundles of efficiency measures with calculated financial costs/benefits for owner to compare each option to the EUI target and baselines established during the earlier simulations. The models can include financial metrics that can be used to assess the impact of efficiency measures on operational costs, return on investment/asset, social cost of carbon and occupant benefits (i.e. indoor environmental quality, etc.). The financial metrics can be used to calculate four levels of value provided to building owners of new, existing and retrofit commercial and residential buildings, including operational benefit, social cost of carbon benefits, asset valuation benefits, and/or occupant benefits.

In some embodiments, the application(s) 522 can transform raw data in to building performance metrics. In other embodiments, the application(s) 522 transmit the raw data to an integrated interface, such as integrated interface 130 shown in FIG. 1 or integrated interface 440 shown in FIG. 4A, to calculate such metrics.

Referring now to FIG. 6 with continuing reference to the foregoing figures, a data acquisition system, generally designated as 600, is shown. The data acquisition system 600 can be configured and implemented as the virtual data acquisition component 110 and/or the physical data acquisition component 120 shown in FIGS. 1A-1B. Similarly, the data acquisition system 600 can be configured and implemented as the virtual data acquisition device 410 and/or to the physical data acquisition device 430 shown in FIGS. 4A-4B.

The data acquisition system 600 can include various components and/or sensor assemblies that can acquire physical data from a building. These components and/or sensor assemblies can include automatic temperature controls 610, a building automation system 612, IEQ/IAQ sensors 614, manual entry devices 616, personal data monitors 618, utility meters 620, and weather stations 622. These components and/or sensor assemblies can communicate with a data acquisition device 624. In this exemplary embodiment, the data acquisition device 624 can be the data acquisition device 500 shown in FIG. 5.

In some embodiments, the automatic temperature controls 610, building automation system 612, IEQ/IAQ sensors 614, manual entry devices 616, personal data monitors 618, utility meters 620, and weather stations 622 can connect to data loggers, such as meters and/or JAVA application control engines (JACES), to facilitate communication over the network 420 shown in FIGS. 4A-4B. In this exemplary embodiment, the JACES can include a JACE 8000—Tridium controller operating with Niagara 4 within the Niagara Framework®. Niagara Framework® is a registered trademark of Tridium, Inc., of Richmond, Virginia.

Referring now to FIG. 7 with continuing reference to the foregoing figures, an integrated interface, generally designated by numeral 700, is shown. The integrated interface 700 can be implemented as a computer, a computer system and/or a computing device. The integrated interface 700 can be the integrated interface 130 shown in FIG. 1. In such embodiments, the integrated interface 700 connects, directly, to one or more data acquisition devices 500 shown in FIG. 5 and/or data acquisition systems 600 shown in FIG. 6.

Alternatively, the integrated interface 700 can be the integrated interface 440 shown in FIGS. 4A-4B. In such

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embodiments, the integrated interface 700 connects to one or more data acquisition devices 500 shown in FIG. 5 and/or data acquisition systems 600 shown in FIG. 6 over a network, such as network 420 shown in FIG. 4A.

The integrated interface 700 can include an assessment platform 710, a processor 712, memory 714, a screen 716, an input device 718, and an output device 720. The integrated interface 700 can host one or more application(s) 722 that include code 724 for enabling the dashboard 450 shown in FIG. 4B.

The integrated interface 700 can include an API 726 that can connect to the energy computation server 470 shown in FIGS. 4A-4B over the network 420. The API 726 is essentially identical to the API 744 shown in FIGS. 4A-4B.

The processor 712 can performing tasks such as signal coding, data processing, input/output processing, power control, and/or other functions. Memory 714 can be used for storing data and/or code for running assessment platform 710 and/or application(s) 722. Example data can include web pages, text, images, sound files, video data, or other data to be sent to and/or received from one or more network servers or other devices via one or more wired and/or wireless networks.

The assessment platform 710, the processor 712, memory 714, and/or the application(s) 722 can cooperate to utilize screen 716 and/or to communicate with input device 718 and/or output device 720.

The assessment platform 710 can utilize the application 722 obtain a virtual building performance model from the virtual data acquisition component 110 shown in FIG. 1 and/or the virtual data acquisition device 410 shown in FIGS. 4A-4B. The assessment platform 710 can utilize the application 722 obtain physical building performance data from the physical data acquisition component 120 shown in FIG. 1 and/or the physical data acquisition device 430 shown in FIGS. 4A-4B. The assessment platform 710 can enable the application 722 to enable comparison of predicted building performance metrics with trended building performance metrics.

The assessment platform 710 can configure and implement the dashboard 450, shown in FIGS. 4A-4B, which can include the landing page 452, the energy use metrics tab 454, the consumption tab 456, and the sensor tab 458. The dashboard 450 can generate visual displays that are based on the predicted building performance metrics and/or the actual building performance metrics to populate the landing page 452, the energy use metrics tab 454, the consumption tab 456, and the sensor tab 458.

The integrated interface 700 can communicate API 726, which can communicate with the portfolio manager application 472 shown in FIGS. 4A-4B to obtain an energy use metric therefrom. The application 722 can utilize the energy use metric to populate the energy use metrics tab 454 shown in FIGS. 4A-4B.

As shown in FIG. 7, the integrated interface 700 can include an analytics platform 728 for processing one or more data streams of trended/metered data parameters. The analytics platform 728 can receive raw data streams and convert the data streams into an analytic data product. The analytic data product can be more informed and advanced. In some embodiments, the analytic data product is based on simulated data.

The use of the analytics platform 728 provides a simulated data product that is informed and advanced. The resulting analytic data products can provide a greater level of insights provided into building performance that can be applied immediately to existing problems, situations or

queries. Exemplary analytics include Fault Detection & Diagnostics (FD&D), Condition-based Maintenance (CBM) and Whole Building Metrics to calculate Site EUI, Source EUI, Total Costs and Greenhouse Gas Emissions.

The analytics platform **728** can include: the SkySpark software analytics platform by SkyFoundry of Glen Allen, Virginia; the Niagara Analytics by Tridium, Inc. of Richmond, Virginia; AWS Analytics by Amazon, Inc. of Seattle, Washington; Google Analytics by Alphabet, Inc. of Mountain View, California; and Microsoft Azure by Microsoft Corp. of Redmond, Washington. The analytics platform **728** can be a proprietary analytics platform, such as a proprietary analytics platform by Honeywell of Charlotte, NC, Siemens of Munich, Germany, Johnson Controls of Cork, Ireland, Automated Logic Corp. of Kennesaw, Georgia, or Schneider Electric of Rueil-Malmaison, France. The analytics platform **728** can be a digital twin enabled through ThoughtWire of Toronto, Ontario (Canada) or Willow, Inc. of Sydney, Australia.

Exemplary Dashboard Output

FIGS. **8A-8F** illustrate exemplary output for the dashboard **450** shown in FIGS. **4A-4B**. It is to be appreciated that the output can be displayed on the client device screen **462** shown in FIGS. **4A-4B**, the screen **518** shown in FIG. **5**, the integrated interface screen **718** shown in FIG. **7** and/or another other similar screen or display device suitable for showing computer generated output.

Referring to FIG. **8A** with continuing reference to the foregoing figures, a landing page **800** is shown. The landing page **800** provides a building owner with key building performance indicators for various portfolios of connected building. The landing page **800** is essentially identical to the landing page **452** shown in FIGS. **4A-4B**.

The landing page **800** includes a drop-down list **802** of building portfolios and/or buildings and a pair of maps **804-806** that can display building icons **808-810** that can represent buildings within a particular building portfolio. The landing page **800** can also display various indicia **812** of building statistics for a particular building, building portfolio, or building project.

The landing page **800** provides building owners with the ability to click onto the building icons **808-810** to obtain various key performance metrics, indicators, sensor outputs, and/or measurements for the buildings that are associated therewith. In this exemplary embodiment, the landing page **800** includes a list **814** of key indicators/metrics for the building associated with building icon **808** and a list **816** of key indicators/metrics for the building associated with building icon **810**. The list **814** includes indicators for site EUI **818**, source EUI **820**, cost **822**, and carbon (i.e., greenhouse gasses) **824** for building icon **808**. The list **816** includes metrics for site EUI **826**, source EUI **828**, cost **830**, and carbon emissions (i.e., greenhouse gas emissions) **832** for building icon **810**.

The landing page **800** includes links **834** to other tabs that are similar to the energy use metrics tab **454**, the consumption tab **456**, and/or the sensor tab **458** shown in FIGS. **4A-4B**.

Referring to FIG. **8B** with continuing reference to the foregoing figures, an energy use metrics tab **836** is shown. The energy use metrics tab **836** provides building owners with the ability to perform an energy use intensity analysis. The energy use metrics tab **836** can display property details and energy use metrics for the current year and for past

years. The energy use metrics tab **836** is essentially identical to the energy use metrics tab **454** shown in FIGS. **4A-4B**.

The energy use metrics tab **836** can display a property icon **838** that can represent a building and/or a portfolio of buildings, a description **840** of the property, and a table **842** that displays various energy use metrics measured at various time embodiments. In this exemplary embodiment, the energy use metrics include EnergyStar® scores **844**, site EUI **846**, source EUI **848**, greenhouse gas emissions **850**, and total utilities costs **852**.

The energy use metrics tab **836** can obtain certain energy use metrics from the energy computation server **470** shown in FIGS. **4A-4B**. The energy computation server **470** enables the energy portfolio manager application **472** to provide automatic EnergyStar® metrics via an EnergyStar® driver.

Referring to FIG. **8C** with continuing reference to the foregoing figures, a consumption tab **854** is shown. The consumption tab **854** can be a utilities consumption tab that provides real-time consumption data from attached metering for a particular building. The consumption tab **854** is essentially identical to the consumption tab **456** shown in FIGS. **4A-4B**.

The consumption tab **854** has the ability to display trended consumption indicia **856** and virtual consumption indicia **858**. The consumption tab **854** displays the data in a bar graph format with an x-axis **860** directed to time and a y-axis **862** directed to energy consumption. In this exemplary embodiment, the time is measured in increments of days and energy consumption is measured in kilowatt hours.

The consumption tab **854** can support all connection types and protocols. The consumption tab **854** can display trending information that can be combined with model trend data to give building owners the ability to control building usage. This information can be used to identify issues ranging from inefficient systems to buildings that are being operated outside of designed/commissioned parameters. The consumption tab **854** can support all sources of utilities and renewables, such as cogeneration systems, district energy chilled-water, district energy steam, electric, gas, potable water, renewable geothermal, renewable photovoltaics, renewable wind, and/or any other source.

Referring to FIGS. **8D-8E** with continuing reference to the foregoing figures, a sensor tab **864** is shown. The sensor tab **864** can display trended output from the physical data acquisition component **120** shown in FIG. **1** and/or the physical data acquisition device **430** shown in FIGS. **4A-4B**. The sensor tab **864** is essentially identical to the sensor tab **458** shown in FIGS. **4A-4B**.

The sensor tab **864** can include a pair of graphs **866-868** that can illustrate real-time temperature, humidity and indoor environmental quality data output. The sensor tab **864** can also include a graphical representation of a floor plan **870**.

The sensor tab **864** can support multiple connection types and protocols to provide trending information relating to sensor output. This trending information can be combined with model trending data to give building owners the ability to control building environmental conditions and air quality, such as ambient lighting, biometrics, CO₂, CO, circadian lighting, formaldehyde, humidity, internally generated noise, Pb, O₃, PM_{2.5}, PM₁₀, Rn, sound pressure level, temperature, VOCs, water quality and/or other environmental factors.

The trending information can be used to produce other charts that illustrate comfort zones or areas for other sensor output. For example, the sensor tab **864** can display a dynamic psychrometric chart **872** that can be used to illus-

trate thermal comfort by identifying areas or zones in which temperature and humidity are at comfortable levels. Specifically, in this exemplary embodiment, the psychrometric chart **872** ties together temperature and humidity levels over time to determine whether the intersections of these three variables fall within certain temperature-humidity comfort zones. The psychrometric chart **872** can be displayed on a thermal comfort meter.

The psychrometric chart **872** can include trended data, predicted data, and/or data produced through the analytics platform **728** shown in FIG. 7, which can be based on predicted data and/or trended data. The comfort zone can be determined using any suitable method, such as the graphic comfort zone method, the analytical comfort zone method, or the elevated air speed comfort zone method.

In other embodiments, the sensor tab **864** can display charts that include other combinations of sensor output to identify other comfort zones. The charts can be used to assess visual comfort, auditory comfort, respiratory comfort or other similar types of comfort. The sensor tab **864** can function as a thermal comfort meter.

An exemplary psychrometric chart **874**, which is similar to the psychrometric chart **872** of FIG. 8D, is shown in more detail in FIG. 8E. The psychrometric chart **874** can include a plot of temperature along an x-axis **876** and humidity along a y-axis **878**. The psychrometric chart **874** can include a comfort zone **880** to illustrate when trended data and/or simulated data is expected to provide a comfortable environment for human occupants.

The psychrometric chart **874** can be used to monitor building performance, to monitor the comfort index, and/or to use simulated data to interrogate building performance. Some data points **882** can fall out of the comfort zone **880**. Other data points **884** can fall within the comfort zone **880**. The data points **882-884** can represent trended data and/or simulated data.

The information provided by the psychrometric chart **874** can be used by design and engineering teams as a tool to change building operations through changes in design or by modifying equipment.

It should be understood that a building will not always operate within the comfort zone **880**. In some instances, such as when a building not expected to be occupied, the building can operate outside of the comfort zone **880**.

The temperature measurements shown in the x-axis **876** can represent air temperature or radiant temperature. It should be further understood that the psychrometric chart **874** can be used to illustrate other variables or parameters, such as clothing insulation, metabolic rate, and air speed. Additionally, humidity can be illustrated on the x-axis **876** and temperature can be illustrated on the y-axis **878**.

Now referring to FIG. 8F with continuing reference to the foregoing figures, a tab, generally designated by the numeral **886**, that includes a campus **888** having a plurality of buildings **890** is shown. The tab **886** can be populated, automatically, with trended data **892** and simulated data **894** for one or more of the buildings **890**.

The trended data **892** and simulated data **894** are populated using a GIS, such as the geographic information system **480** shown in FIG. 4A. In this exemplary embodiment, the trended data **892** and simulated data **894** are populated using ArcGIS, which represents a suite of products such as ArcMAP and Arc Catalog, which provides software tools for visualizing and analyzing data. ArcMAP can be used to display and to explore ArcGIS datasets.

Exemplary Processes

Referring to FIGS. 9A-9B with continuing reference to the foregoing figures, a method **900** is illustrated as an

embodiment of an exemplary process for assessing building performance in accordance with features of the described subject matter. Method **900**, or portions thereof, can be performed by one or more computing devices, a computer system, computer-executable instructions, software, hardware, firmware or a combination thereof in various embodiments. For example, method **900** can be performed by system **100** shown in FIG. 1, system **400** shown in FIGS. 4A-4B, or any other suitable system.

At **901**, an integrated interface receives a building performance model having a plurality of predicted building performance metrics. In this exemplary embodiment, the integrated interface can be integrated interface **130** shown in FIG. 1, integrated interface **440** shown in FIGS. 4A-4B, and/or integrated interface **700** shown in FIG. 7. The building performance model can be received from virtual data acquisition component **110** shown in FIG. 1, virtual data acquisition device **410** shown in FIGS. 4A-4B and/or data acquisition device **500** shown in FIG. 5.

At **902**, the integrated interface receives a plurality of trended building performance metrics. In this exemplary embodiment, the trended building performance metrics can be received from physical data acquisition component **120** shown in FIG. 1, physical data acquisition device **430** shown in FIGS. 4A-4B and/or data acquisition system **600** shown in FIG. 6.

At **903**, a platform is generated to facilitate the comparison of the predicted building performance metrics with the trended building performance metrics to identify performance gaps. In this exemplary embodiment, the platform can be the building performance assessment platform **132** shown in FIG. 1.

At **904**, the building performance model is modified to produce an enhanced building performance model. In this exemplary embodiment, the building performance model can be modified by virtual data acquisition component **110** shown in FIG. 1, virtual data acquisition device **410** shown in FIGS. 4A-4B and/or data acquisition device **500** shown in FIG. 5.

At **905**, the predicted building performance metrics and the trended building performance metrics are received in essentially the same format to form an integrated data stream. In this exemplary embodiment, the integrated interface **130** can configure and implement the building performance assessment platform **132** to form the integrated data stream.

At **906**, an interrogation based commissioning assistance module is enabled to utilize the integrated data stream. In this exemplary embodiment, the interrogation based commissioning assistance module can be the interrogation-based commissioning assistance module **134** shown in FIG. 1.

At **907**, the integrated interface produces output for a dynamic decision matrix through the dashboard. In this exemplary embodiment, the integrated interface **130** configures and implements the dynamic decision matrix output production module **136** shown in FIG. 1.

At **908**, the integrated interface provides a dynamic display through the dashboard. In this exemplary embodiment, the dynamic display can be produced through the dashboard display **138** shown in FIG. 1.

At **909**, the integrated interface provides output to enable impact assessments for owners for sustainability goals, first costs and long term operating costs through the dashboard. In this exemplary embodiment the dashboard display **138** shown in FIG. 1 can be used to display the output.

At **910**, the integrated interface provides output to enable visual identification of the relationship between trended and

predicted building performance. In this exemplary embodiment the dashboard display **138** shown in FIG. **1** can be used to display the output.

At **911**, the building performance model is modified to produce an enhanced building performance model. In this exemplary embodiment, the integrated interface **130** shown in FIG. **1**, the integrated interface **440** shown in FIGS. **4A-4B**, and/or the integrated interface **700** shown in FIG. **7** can communicate with the virtual data acquisition component **110** shown in FIG. **1**, the virtual data acquisition device **410** shown in FIGS. **4A-4B** and/or the data acquisition device **500** shown in FIG. **5** to produce the enhanced building performance model.

Supported Embodiments

The detailed description provided above in connection with the appended drawings explicitly describes and supports various features of an improved building performance assessment system in accordance with the described subject matter. By way of illustration and not limitation, supported embodiments include a building interrogation system comprising: a virtual data acquisition component for holding a physics-based, building energy model having a plurality of predicted building performance metrics produced through a simulation of expected building performance, a physical data acquisition component for obtaining a plurality of trended building performance metrics during the operation of a building, an integrated interface that receives the plurality of predicted building performance metrics from the virtual data acquisition component and the plurality of trended building performance metrics from the physical data acquisition component to form an integrated data stream having integrated data stream data therein, a display device for displaying a psychometric chart having at least one comfort zone thereon, wherein the integrated interface communicates the integrated data stream data to the display device to populate the psychometric chart.

Supported embodiments include the foregoing system, wherein the integrated data stream data include trended measurements and predicted measurements.

Supported embodiments include any of the foregoing systems, wherein the trended measurements include trended temperature measurements and trended humidity measurements.

Supported embodiments include any of the foregoing systems, wherein the predicted measurements include predicted temperature measurements and predicted humidity measurements.

Supported embodiments include any of the foregoing systems, wherein the trended measurements include trended temperature measurements and trended humidity measurements.

Supported embodiments include any of the foregoing systems, wherein the integrated data stream data include parameters selected from the group consisting of clothing insulation, air temperature, radiant temperature, humidity, metabolic rate, and air speed.

Supported embodiments include any of the foregoing systems, wherein the display device can display one of the parameters on an x-axis on the psychometric chart and another one of the parameters on a y-axis on the psychometric chart.

Supported embodiments include a computer-implemented method, an apparatus, a computer-readable storage medium, and/or means for implementing and/or performing any of the foregoing systems or portions thereof.

Supported embodiments include an apparatus for monitoring building comfort, the apparatus comprising: a virtual data acquisition component for calibrating a virtual building performance model having a plurality of predicted building performance metrics produced through a physics-based simulation of expected building performance, a physical data acquisition system for acquiring physical data from a building, an integrated interface for receiving the plurality of predicted building performance metrics from the virtual data acquisition component and the physical data from the physical data acquisition system to form an integrated data stream having integrated data stream data therein, an output device for displaying a psychometric chart having at least one comfort zone thereon, wherein the integrated interface communicates the integrated data stream data to the output device to populate the psychometric chart.

Supported embodiments include the foregoing apparatus, wherein the integrated data stream data include trended measurements and predicted measurements.

Supported embodiments include any of the foregoing apparatuses, wherein the trended measurements include trended temperature measurements and trended humidity measurements.

Supported embodiments include any of the foregoing apparatuses, wherein the predicted measurements include predicted temperature measurements and predicted humidity measurements.

Supported embodiments include any of the foregoing apparatuses, wherein the trended measurements include trended temperature measurements and trended humidity measurements.

Supported embodiments include any of the foregoing apparatuses, wherein the integrated data stream data include parameters selected from the group consisting of clothing insulation, air temperature, radiant temperature, humidity, metabolic rate, and air speed.

Supported embodiments include any of the foregoing apparatuses, wherein the output device can display one of the parameters on an x-axis on the psychometric chart and another one of the parameters on a y-axis on the psychometric chart.

Supported embodiments include a computer-implemented method, a system, a computer-readable storage medium, and/or means for implementing and/or performing any of the foregoing apparatuses or portions thereof.

Supported embodiments include a computing-implemented method for optimizing building performance, the method comprising: calibrating, with a virtual data acquisition device, a virtual building performance model having a plurality of predicted building performance metrics produced through a physics-based simulation of expected building performance, obtaining, with a physical data acquisition system, physical data from a building, receiving, with an integrated interface, the plurality of predicted building performance metrics from the virtual data acquisition component and the physical data from the physical data acquisition system, forming, with the integrated interface, an integrated data stream having integrated data stream data therein, and generating output for displaying the integrated data stream data on a psychometric chart with a comfort zone thereon.

Supported embodiment include the foregoing computing-implemented method, further comprising: communicating the output from the integrated data stream data to a display device, and displaying, on the display device, the psychometric chart.

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Supported embodiments include any of the foregoing computing-implemented methods, further comprising: displaying, on the display device, the psychometric chart in real-time.

Supported embodiments include any of the foregoing computing-implemented methods, wherein the integrated data stream data includes parameters selected from the group consisting of clothing insulation, air temperature, radiant temperature, humidity, metabolic rate, and air speed, further comprising: displaying, on the display device, one of the parameters on an x-axis on the psychometric chart and another one of the parameters on a y-axis on the psychrometric chart.

Supported embodiments include any of the foregoing computing-implemented methods, further comprising: populating the psychometric chart with the integrated data stream data.

Supported embodiments include any of the foregoing computing-implemented methods, wherein the integrated data stream data include trended measurements and predicted measurements.

Supported embodiments include a system, an apparatus, a computer-readable storage medium, and/or means for implementing and/or performing any of the foregoing computer-implemented methods or portions thereof.

Supported embodiments include a system for assessing building performance for a building, the system comprising: a virtual data acquisition component for holding a physics-based, building energy model having a plurality of predicted building performance metrics produced through a simulation of expected building performance, a physical data acquisition component for obtaining a plurality of trended building performance metrics during the operation of a building, and an integrated interface that receives the plurality of predicted building performance metrics from the virtual data acquisition component and the plurality of trended building performance metrics from the physical data acquisition component to form an integrated data stream, wherein the integrated interface calculates at least one analytic from at least one of the plurality of predicted building performance metrics and at least one of the plurality of trended building performance metrics within the integrated data stream and communicates the at least one analytic as output for a display device.

Supported embodiments include an apparatus for assessing building performance for a building, the apparatus comprising: a virtual data acquisition component for holding a physics-based, building energy model having a plurality of predicted building performance metrics produced through a simulation of expected building performance, a physical data acquisition component for obtaining a plurality of trended building performance metrics during the operation of a building, and an integrated interface that receives the plurality of predicted building performance metrics from the virtual data acquisition component and the plurality of trended building performance metrics from the physical data acquisition component to form an integrated data stream, wherein the integrated interface populates a geodata data structure for a geographic information system with the plurality of predicted building performance metrics and the plurality of trended building performance metrics within the integrated data stream, so that the geographic information system can generate a geographic map having a plurality of geospatial locations with each of the plurality of geospatial locations having at least one of the plurality of predicted building performance metrics and at least one of the plural-

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ity of trended building performance metrics for access by a user of the geographic information system.

Supported embodiments include a system for assessing building performance for a building, the system comprising: a virtual data acquisition component for holding a physics-based, building energy model having a plurality of predicted building performance metrics produced through a simulation of expected building performance, a physical data acquisition component for obtaining a plurality of trended building performance metrics during the operation of a building, and an integrated interface having an analytics platform thereon, wherein the integrated interface receives the plurality of predicted building performance metrics from the virtual data acquisition component and the plurality of trended building performance metrics from the physical data acquisition component, wherein the integrated interface analytics platform converts at least one of the plurality of predicted building performance metrics and the plurality of trended building performance metrics into an analytic data product and forms an integrated data stream that includes the analytic data product, and wherein the integrated interface communicates the integrated data stream to an output device for display to provide a user with the ability to compare the analytic data product with at least one of the plurality of predicted building performance metrics and the trended building performance metrics and to identify the building performance gaps therein.

Supported embodiments include the foregoing system, wherein the plurality of predicted building performance metrics are produced through a simulation of expected building performance.

Supported embodiments include any of the foregoing systems, wherein the plurality of predicted building performance metrics are produced through a simulation of building performance as designed.

Supported embodiments include any of the foregoing systems, wherein the integrated data stream includes multiple data streams for comparison of simulated performance to trended performance.

Supported embodiments include any of the foregoing systems, wherein the integrated data stream provides the ability to validate building performance metrics during a post-construction phase.

Supported embodiments include any of the foregoing systems, the integrated interface communicates the integrated data stream to the output device for display of a single parameter data stream.

Supported embodiments include any of the foregoing systems, the integrated interface communicates the integrated data stream to the output device for display of a combined parameter data stream.

Supported embodiments include any of the foregoing systems, the integrated interface communicates the integrated data stream to the output device for display of a psychrometric chart.

Supported embodiments include any of the foregoing systems, the integrated interface has the ability to communicate the integrated data stream to the output device upon construction of the building.

Supported embodiments include any of the foregoing systems, wherein the integrated interface receives manually-entered data for the physics-based building energy model.

Supported embodiments include any of the foregoing systems, wherein the integrated interface includes an assessment platform for comparing building performance goals, building construction costs and long term operating costs.

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Supported embodiments include any of the foregoing systems, wherein the assessment platform provides the ability to compare building performance goals, building construction costs and long term operating costs for portfolios of buildings to allow building owners to prioritize investments based on returns.

Supported embodiments include any of the foregoing systems, wherein the assessment platform is a building performance assessment platform.

Supported embodiments include any of the foregoing systems, wherein the assessment platform includes a dynamic decision matrix output production module to provide output for a dynamic decision matrix on the output device.

Supported embodiments include any of the foregoing systems, wherein the dynamic decision matrix includes building performance goals, building construction costs, and long-term operational costs for a building.

Supported embodiments include any of the foregoing systems, wherein the assessment platform implements an interrogation-based commissioning module to identify performance corrections.

Supported embodiments include any of the foregoing systems, wherein the physical data acquisition component is selected from the group consisting of a physical data input device and a data transmission device.

Supported embodiments include any of the foregoing systems, wherein the integrated interface produces output that displays the plurality of predicted building performance metrics and the plurality of physical building performance metrics as a function of time to illustrate the performance gaps.

Supported embodiments include a computer-implemented method, an apparatus, a computer-readable storage medium, and/or means for implementing and/or performing any of the foregoing systems or portions thereof.

Supported embodiments include a system for assessing building performance of a building, the system comprising: a virtual data acquisition device, a physical data acquisition device, an integrated interface having an analytics platform thereon, a client device, and a network connecting the virtual data acquisition device to the physical data acquisition device, the integrated interface, and the client device, wherein the virtual data acquisition device holds a building energy model having a plurality of predicted building performance metrics for the building with the building energy model being a design model, wherein the physical data acquisition device obtains a plurality of trended building performance metrics for the building to transform the building energy model from a design model into an operational model, wherein the integrated interface receives the plurality of predicted building performance metrics from the virtual data acquisition component and the plurality of trended building performance metrics from the physical data acquisition component, wherein the integrated interface analytics platform converts at least one of the plurality of predicted building performance metrics and the plurality of trended building performance metrics into an analytic data product and forms an integrated data stream that includes the analytic data product, and wherein the integrated interface communicates the integrated data stream to a client device for display to provide a user with the ability to compare the analytic data product with at least one of the plurality of predicted building performance metrics and the trended building performance metrics and to identify the building performance gaps therein.

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Supported embodiments include the foregoing system, wherein the virtual data acquisition device stores the building energy model.

Supported embodiments include any of the foregoing systems, wherein the plurality of predicted building performance metrics are produced through a simulation of expected building performance.

Supported embodiments include any of the foregoing systems, wherein the plurality of predicted building performance metrics are produced through a simulation of building performance as designed.

Supported embodiments include any of the foregoing systems, wherein the integrated data stream includes multiple data streams for comparison of simulated performance to trended performance.

Supported embodiments include any of the foregoing systems, wherein the integrated data stream provides the ability to validate building performance metrics during a post-construction phase.

Supported embodiments include any of the foregoing systems, the integrated interface communicates the integrated data stream to the output device for display of a single parameter data stream.

Supported embodiments include any of the foregoing systems, the integrated interface communicates the integrated data stream to the output device for display of a combined parameter data stream.

Supported embodiments include any of the foregoing systems, the integrated interface communicates the integrated data stream to the output device for display of a psychrometric chart.

Supported embodiments include any of the foregoing systems, the integrated interface has the ability to communicate the integrated data stream to the output device upon construction of the building.

Supported embodiments include any of the foregoing systems, wherein the integrated interface receives manually-entered data for the physics-based building energy model.

Supported embodiments include any of the foregoing systems, wherein the integrated interface includes an assessment platform for comparing building performance goals, building construction costs and long term operating costs.

Supported embodiments include any of the foregoing systems, wherein the assessment platform provides the ability to compare building performance goals, building construction costs and long term operating costs for portfolios of buildings to allow building owners to prioritize investments based on returns.

Supported embodiments include any of the foregoing systems, wherein the assessment platform is a building performance assessment platform.

Supported embodiments include any of the foregoing systems, wherein the assessment platform includes a dynamic decision matrix output production module to provide output for a dynamic decision matrix on the output device.

Supported embodiments include any of the foregoing systems, wherein the dynamic decision matrix includes building performance goals, building construction costs, and long-term operational costs for a building.

Supported embodiments include any of the foregoing systems, wherein the assessment platform implements an interrogation-based commissioning module to identify performance corrections.

Supported embodiments include any of the foregoing systems, wherein the physical data acquisition component is

selected from the group consisting of a physical data input device and a data transmission device.

Supported embodiments include any of the foregoing systems, wherein the integrated interface produces output that displays the plurality of predicted building performance metrics and the plurality of physical building performance metrics as a function of time to illustrate the performance gaps.

Supported embodiments include a computer-implemented method, an apparatus, a computer-readable storage medium, and/or means for implementing and/or performing any of the foregoing systems or portions thereof.

Supported embodiments include an arrangement configured to provide for the assessment of building performance for a building, the arrangement comprising: an output to a display screen configured to display an integrated data stream providing a user with the ability to compare a plurality of predicted building performance metrics to a plurality of trended building performance metrics and to identify building performance gaps therein, memory for storing an integrated data stream that includes the plurality of predicted building performance metrics after the plurality of predicted building performance metrics are received from a virtual data acquisition component and the plurality of trended building performance metrics after the plurality of trended building performance metrics are received from a physical data acquisition component, and at least one processor coupled to said memory and said output and operatively configured to process the integrated data stream for display on the display screen, wherein the plurality of predicted building performance metrics are obtained from a physics-based, building energy model having a plurality of predicted building performance metrics produced through a simulation of expected building performance, and wherein the plurality of trended building performance metrics are obtained during the operation of the building.

Supported embodiments include the foregoing arrangement, wherein the processor is coupled to an integrated interface having an analytics platform for converting at least one of the plurality of predicted building performance metrics and the plurality of trended building performance metrics into an analytic data product.

Supported embodiments include any of the foregoing arrangements, wherein the integrated data stream that includes the analytic data product.

Supported embodiments include any of the foregoing arrangements, wherein the display provides a user with the ability to compare the analytic data product with at least one of the plurality of predicted building performance metrics and the trended building performance metrics and to identify the building performance gaps therein.

Supported embodiments include any of the foregoing arrangements, further comprising: a client device having display screen, memory, and the at least one processor.

Supported embodiments include a computer-implemented method, a system, an apparatus, a computer-readable storage medium, and/or means for implementing and/or performing any of the foregoing arrangements or portions thereof.

Supported embodiments include a computer-implemented method, the method comprising: receiving, at an integrated interface having an analytics platform, a plurality of predicted building performance metrics from a virtual data acquisition component and a plurality of trended building performance metrics from a physical data acquisition component; converting, with the analytics platform, at least one of the plurality of predicted building performance metrics and the plurality of trended building performance metrics

into an analytic data product; forming, with the analytics platform, an integrated data stream that includes the analytic data product; and communicating, with the integrated interface, the integrated data stream to an output device for display to provide a user with the ability to compare the analytic data product with at least one of the plurality of predicted building performance metrics and the trended building performance metrics and to identify the building performance gaps therein.

Supported embodiments include a computer program product produced through the foregoing method.

Supported embodiments can provide various attendant and/or technical advantages in terms of an instrumentality that can interrogate buildings and monitor comfort on a real-time basis. The instrumentality can calculate analytics from trended data and predicted data. The trended data and the predicted can populate geodata structures for display on geographic maps in real-time.

The detailed description provided above in connection with the appended drawings is intended as a description of examples and is not intended to represent the only forms in which the present examples can be constructed or utilized.

It is to be understood that the configurations and/or approaches described herein are exemplary in nature, and that the described embodiments, implementations and/or examples are not to be considered in a limiting sense, because numerous variations are possible. The specific processes or methods described herein can represent one or more of any number of processing strategies.

As such, various operations illustrated and/or described can be performed in the sequence illustrated and/or described, in other sequences, in parallel, or omitted. For example, it should be understood that an embodiment is contemplated in which a building monitoring system calculates or determines various building performance metrics by obtaining those metrics from an energy computation server. Another exemplary embodiment is contemplated in which a building model generating system resides on the same computer, computing device, and/or computer system as a building monitoring system.

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are presented as example forms of implementing the claims.

What is claimed is:

1. An apparatus for optimizing building performance, the apparatus comprising:

- a first computing device hosting a virtual data acquisition component for holding a virtual building performance model having predicted building performance metrics that were produced through a physics-based simulation of expected building performance;
 - a second computing device hosting a physical data acquisition component having sensors for acquiring trended building performance metrics from a building;
 - a third computing device hosting an integrated interface with an analytics platform, having an input device, and connecting to the first computing device and to the second computing device over a network; and
 - a display device for displaying a dashboard thereon connecting to the third computing device;
- wherein the physical data acquisition component acquires trended building performance metrics from the building through the sensors and transmits the trended

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building performance metrics from the second computing device and to the third computing device over the network;

wherein the integrated interface receives the trended building performance metrics from the third computing device over the network and uses the trended building performance metrics to modify the virtual building performance model, on a continuous basis, to transform the virtual building performance model into a calibrated model;

wherein the integrated interface analytics platform converts at least one of the predicted building performance metrics and the trended building performance metrics into an analytic data product and forms an integrated data stream that includes the analytic data product;

wherein the integrated interface communicates the integrated data stream to an output device for display thereon; and

wherein the third computing device receives input through the input device for modifying the display of the integrated data stream on the display device in real-time.

2. The apparatus of claim 1, wherein the predicted building performance metrics are produced through a simulation of expected building performance.

3. The apparatus of claim 1, wherein the predicted building performance metrics are produced through a simulation of building performance as designed.

4. The apparatus of claim 1, wherein the integrated data stream includes multiple data streams for comparison of simulated performance to trended performance.

5. The apparatus of claim 1, wherein the integrated interface communicates the integrated data stream to the output device for display of a single parameter data stream.

6. The apparatus of claim 1, wherein the integrated interface communicates the integrated data stream to the output device for display of a combined parameter data stream.

7. The apparatus of claim 1, wherein the integrated interface communicates the integrated data stream to the output device for display of a psychrometric chart.

8. The apparatus of claim 1, wherein the integrated interface receives manually-entered data for the physics-based building energy model through the input device.

9. The apparatus of claim 1, wherein the integrated interface includes a dynamic decision matrix for comparing building performance goals, building construction costs and long term operating costs.

10. The apparatus of claim 1, wherein the integrated interface includes an interrogation-based commissioning module to identify performance gaps.

11. The apparatus of claim 1, wherein the physical data acquisition component is selected from the group consisting of a physical data input device and a data transmission device.

12. The apparatus of claim 1, wherein the integrated interface produces output that displays the predicted build-

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ing performance metrics and the physical building performance metrics as a function of time to illustrate the performance gaps.

13. A method for optimizing building performance, the method comprising:

hosting, on a first computing device, a virtual data acquisition component having a virtual building performance model with predicted building performance metrics that were produced through a physics-based simulation of expected building performance;

hosting, on a second computing device, a physical data acquisition component having sensors for acquiring trended building performance metrics from a building; connecting, over a network, the first computing device to the second computing device with an integrated interface with an analytics platform thereon;

acquiring, with the physical data acquisition component, trended building performance metrics from the building through the sensors;

modifying the virtual building performance model with the trended building performance metrics to transform the virtual building performance model into a calibrated building performance model with modified predicted building performance metrics;

converting, with the integrated interface analytics platform, the modified predicted building performance metrics into an analytic data product;

forming an integrated data stream from the analytic data product; and

communicating the integrated data stream to an output device for display thereon.

14. The method of claim 13, further comprising: receiving, through an input device connected to the integrated interface, input; and modifying the integrated data stream based upon the input in real-time.

15. The method of claim 13, further comprising: displaying a psychrometric chart on the output device.

16. The method of claim 13, further comprising: displaying a dynamic decision matrix for comparing building performance goals, building construction costs and long-term operating costs on the output device.

17. The method of claim 13, wherein the integrated interface includes an interrogation-based commissioning module.

18. The method of claim 13, further comprising: receiving predicted building performance metrics that were produced through a simulation of expected building performance.

19. The method of claim 13, further comprising: receiving predicted building performance metrics predicted building performance metrics that were produced through a simulation of building performance as designed.

20. The method of claim 13, further comprising: forming multiple data streams for comparison of simulated performance to trended performance.

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